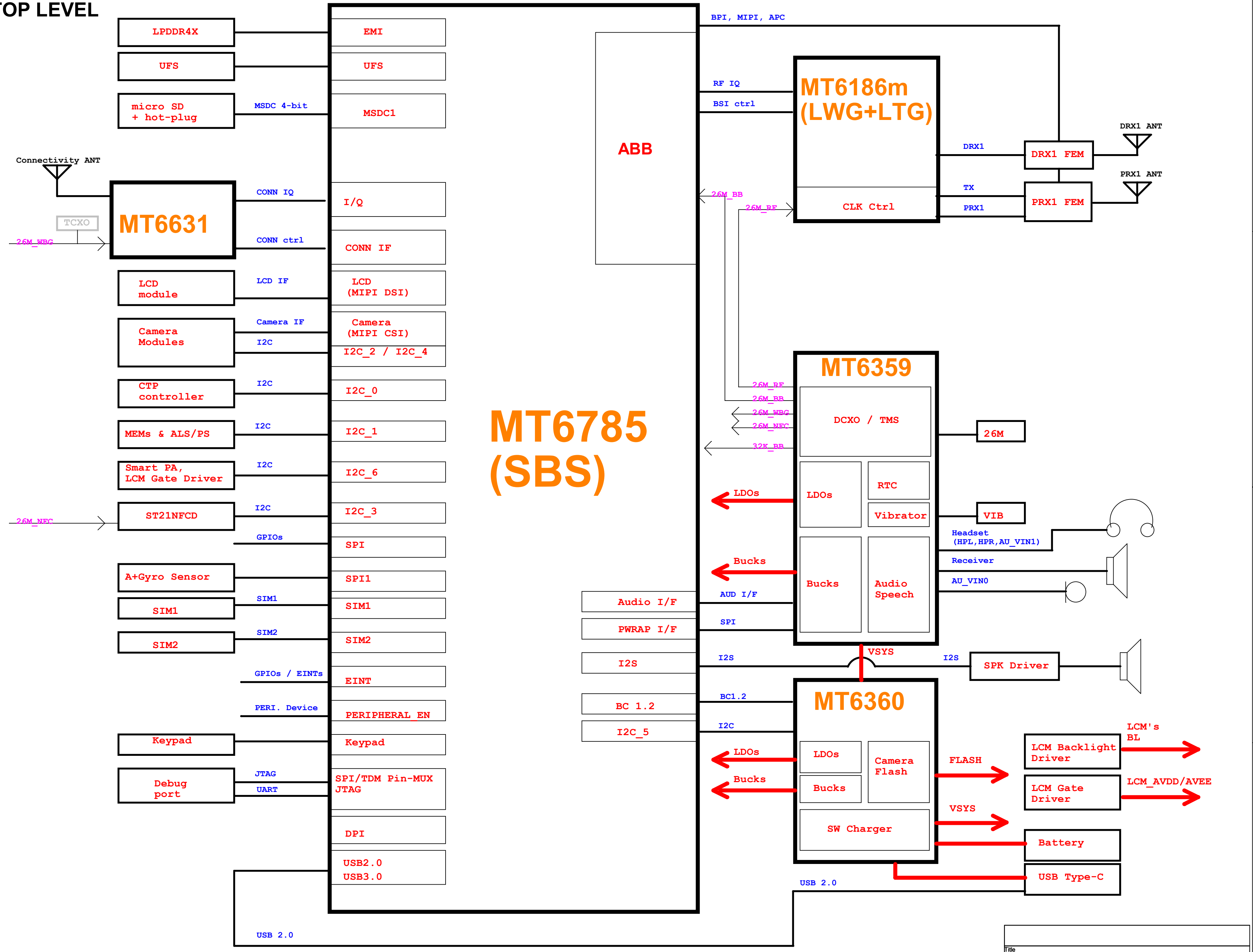
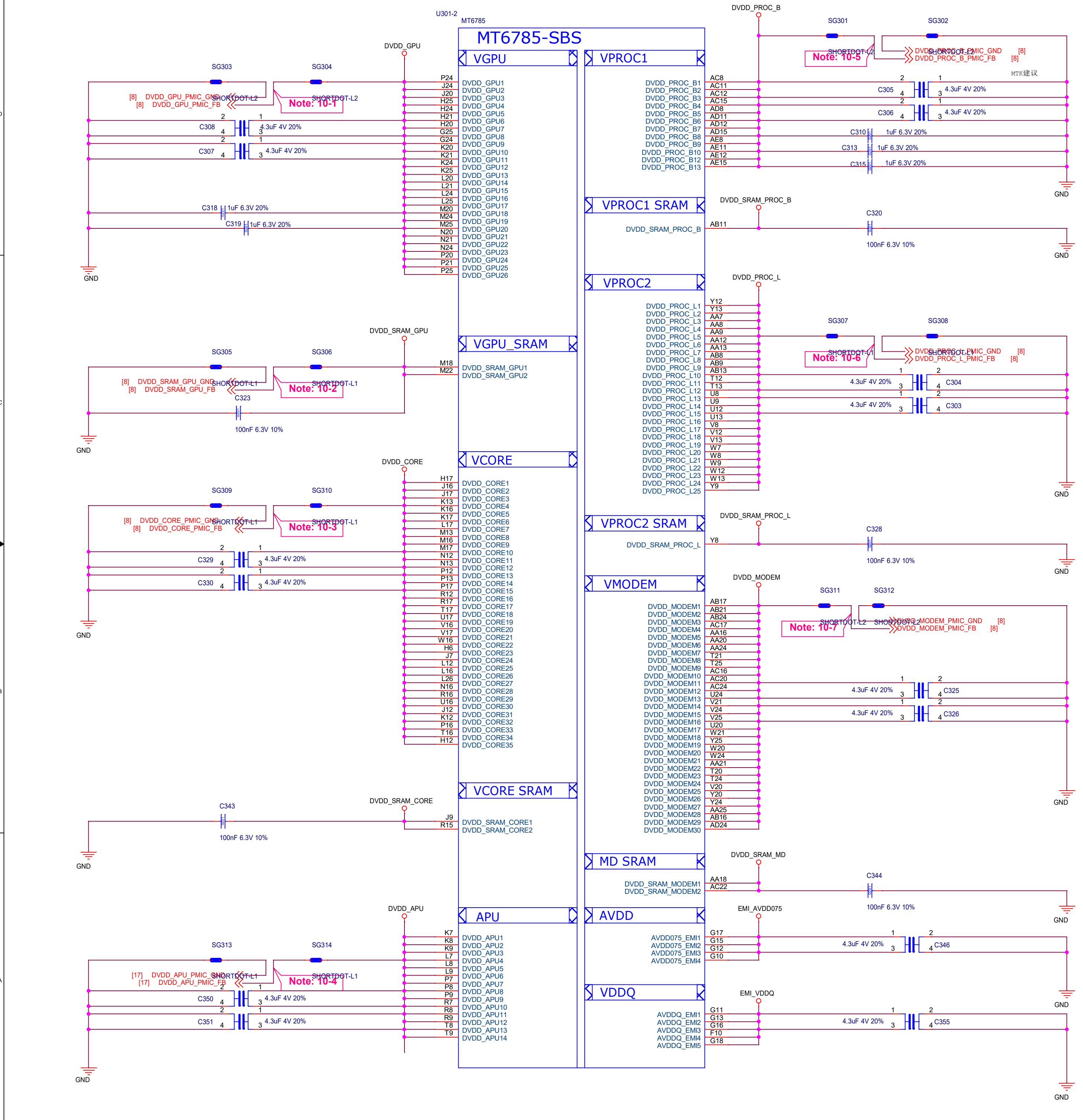
Project : MT6785 LPDDR4X
REF_SCH TOP LEVEL



Title
01_Block_Diagram
Size C
MTK Confidential
Date: Monday, December 14, 2020 Sheet 1 of 55



Schematic design notice of "10_BB_POWER_PDN" page.

Note 10-1:	Differential pair of DVDD_GPU remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-2:	Differential pair of DVDD_SRAM_GPU remote sense must be close to 22uF due to SRAM application. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-3:	Differential pair of DVDD_CORE remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-4:	Differential pair of DVDD_APU remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-5:	Differential pair of DVDD_PROC_B remote sense must be close to BB's ball Remote sense trace with GND shielding to PMIC (Differential)
Note 10-6:	Differential pair of DVDD_PROC_L remote sense must be close to BB's ball Remote sense trace with GND shielding to PMIC (Differential)
Note 10-7:	Differential pair of DVDD_MODEM remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)

Title: 03_BB_POWER_PDN

Size: C

MTK Confidential

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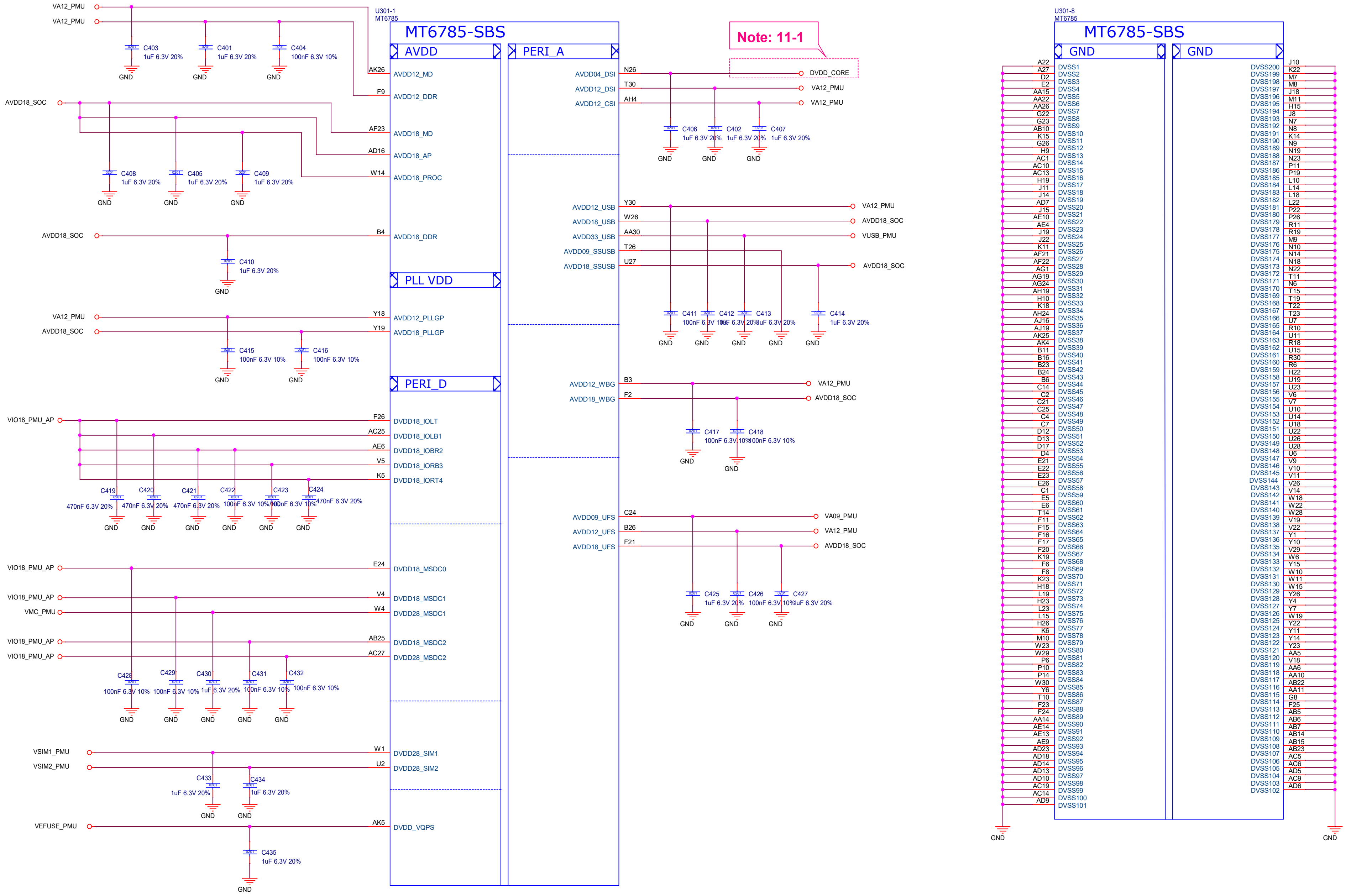

Schematic design notice of "11_BB_POWER_IO" page.

Note 11-1: For D-PHY application, connect MIPI Power domain AVDD04_DSI to DVDD_CORE.
For C-PHY application, connect MIPI Power domain AVDD04_DSI to VA12_PMU.

Title	04_BB_POWER_IO
Size C	MTK Confidential
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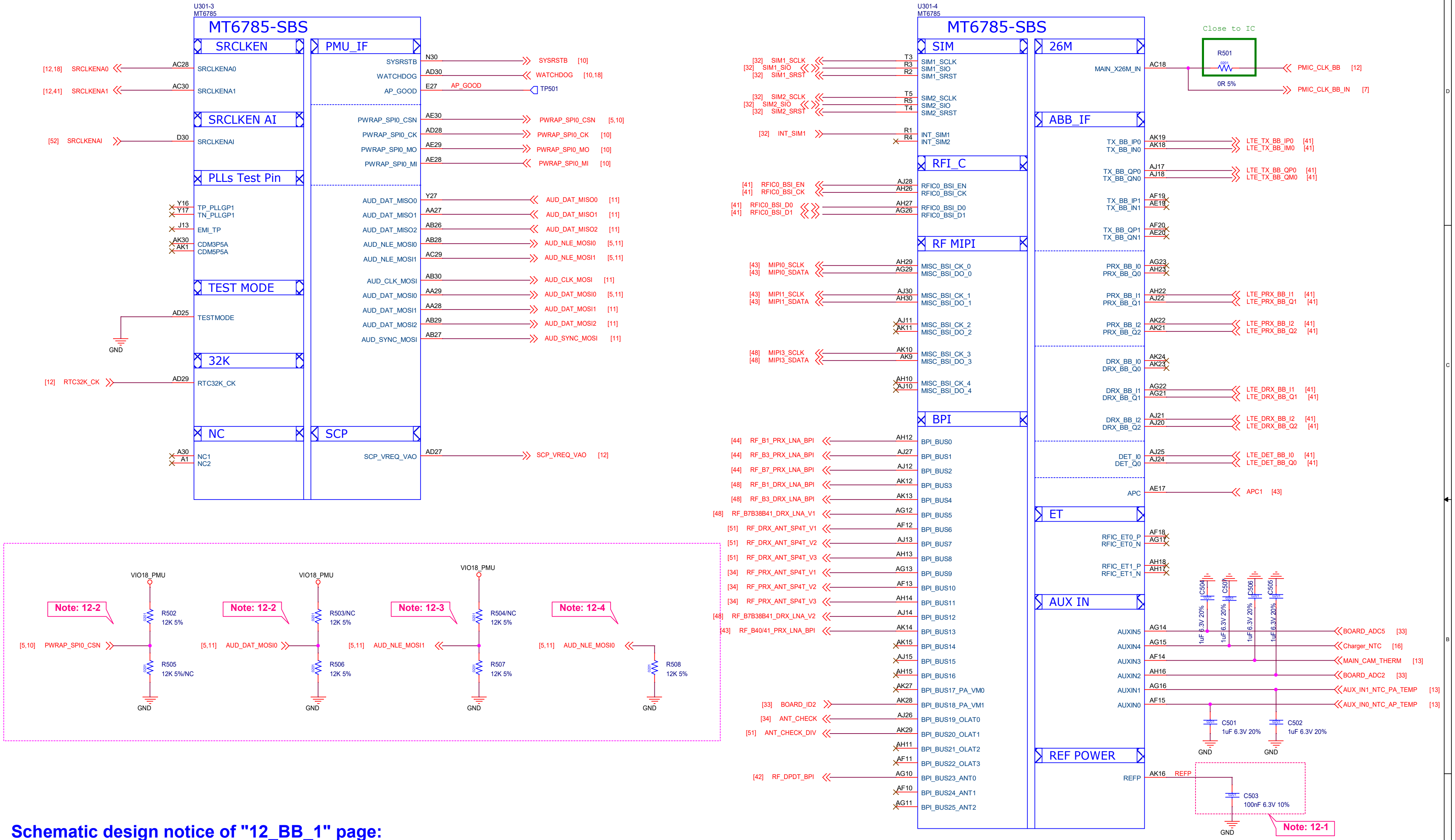

Schematic design notice of "12_BB_1" page:

Note 12-1: The de-coupling cap. for REFP (AK16 ball) have to be placed as close to BB as possible.

Note 12-2: PWRAP_SPI0_CSN and AUD_DAT_MOSI are JTAG feature in bootstrap.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP_JTAG	IO_JTAG
HI	LO	N/A	N/A
HI	HI (by external PU)	SPI_CSB/SPI_CLK/ SPI_MO/SPI_MI/EINT3	N/A
LO (by external PD)	LO	SPI_CSB/SPI_CLK/ SPI_MO/SPI_MI/EINT3	TDM_LRCK/TDM_BCK/TDM_MCLK/ TDM_DATA0/TDM_DATA1
LO (by external PD)	HI (by external PU)	MSDC1_CLK/CMD/ DAT0/DAT1/DAT2	N/A

Note 12-3: AUD_NLE_MOSI1 is USB JTAG feature in bootstrap.

AUD_NLE_MOSI1	USB JTAG
LO	USB 2.0
HI (by external PU)	DP/DM output JTAG

Note 12-4: AUD_NLE_MOSI0 is storage booting feature in bootstrap.

AUD_NLE_MOSI0	Storage Booting
LO	UFS only
HI (by external PU)	eMMC only

Title 05_BB_1

Size C

MTK Confidential

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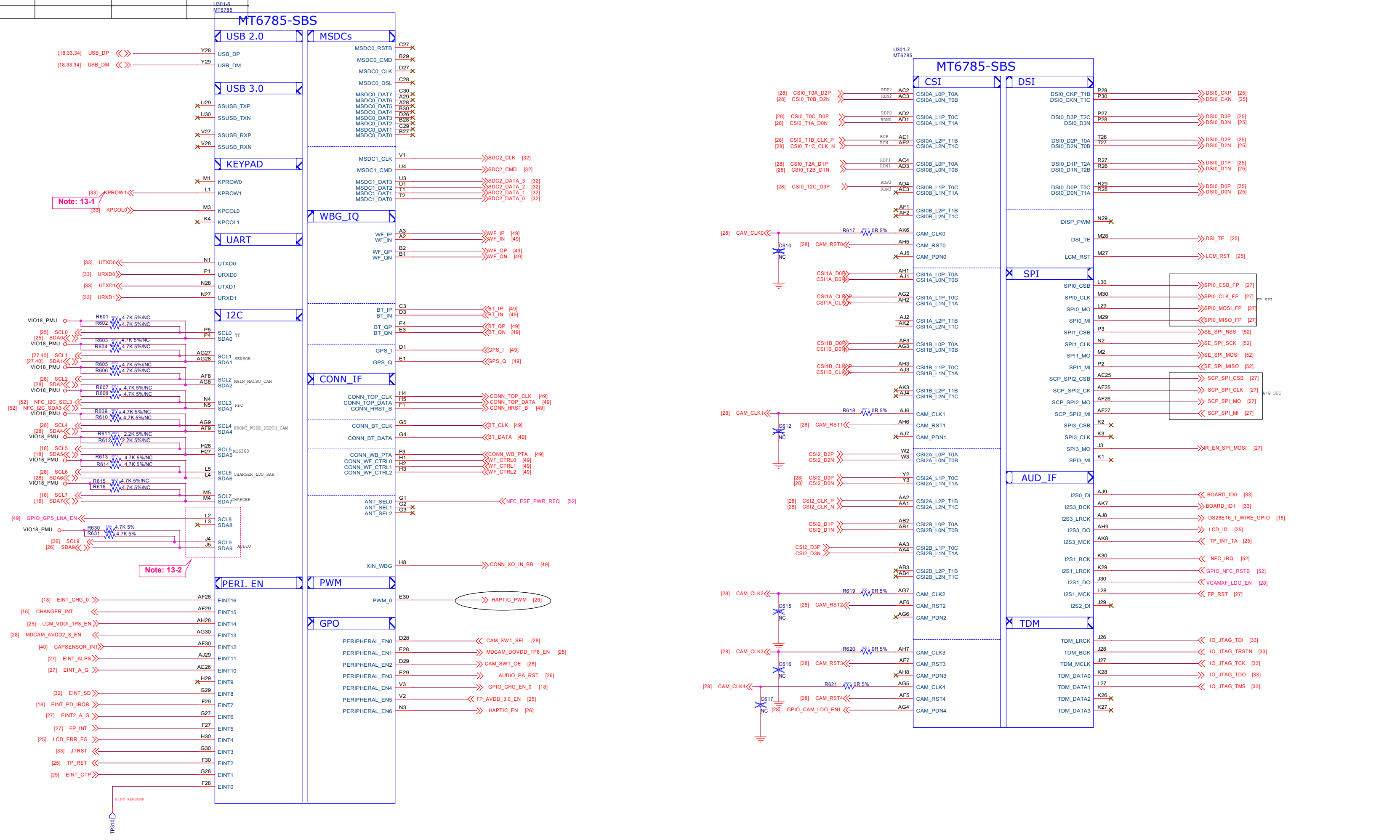
I2C	Device1	Device2	Device3	Device4	Device5	Device6
0	CTP					
1	TSL2540	BU27030NUC	AK09918C	Sar_SENSOR		
2	Main_CAM	Macro_CAM				
3	NFC ST21					
4	Front_CAM	Depth_CAM				
5	MT6360					
6	WL2866D-10/TR	Wide_CAM				
7	SC8551					
9	AW87559FCR	AW87389FCR				



Schematic design notice of "06_BB_2" page.

Note 13-1: Reserved for fast META mode detection

Note 13-2: For (I2C-8, I2C-9), must have pull up resistor 4.7K ohm for standard mode and fast mode; 2.2K ohm for fast mode plus and high speed mode.

Title: 06_BB_2

Size: D

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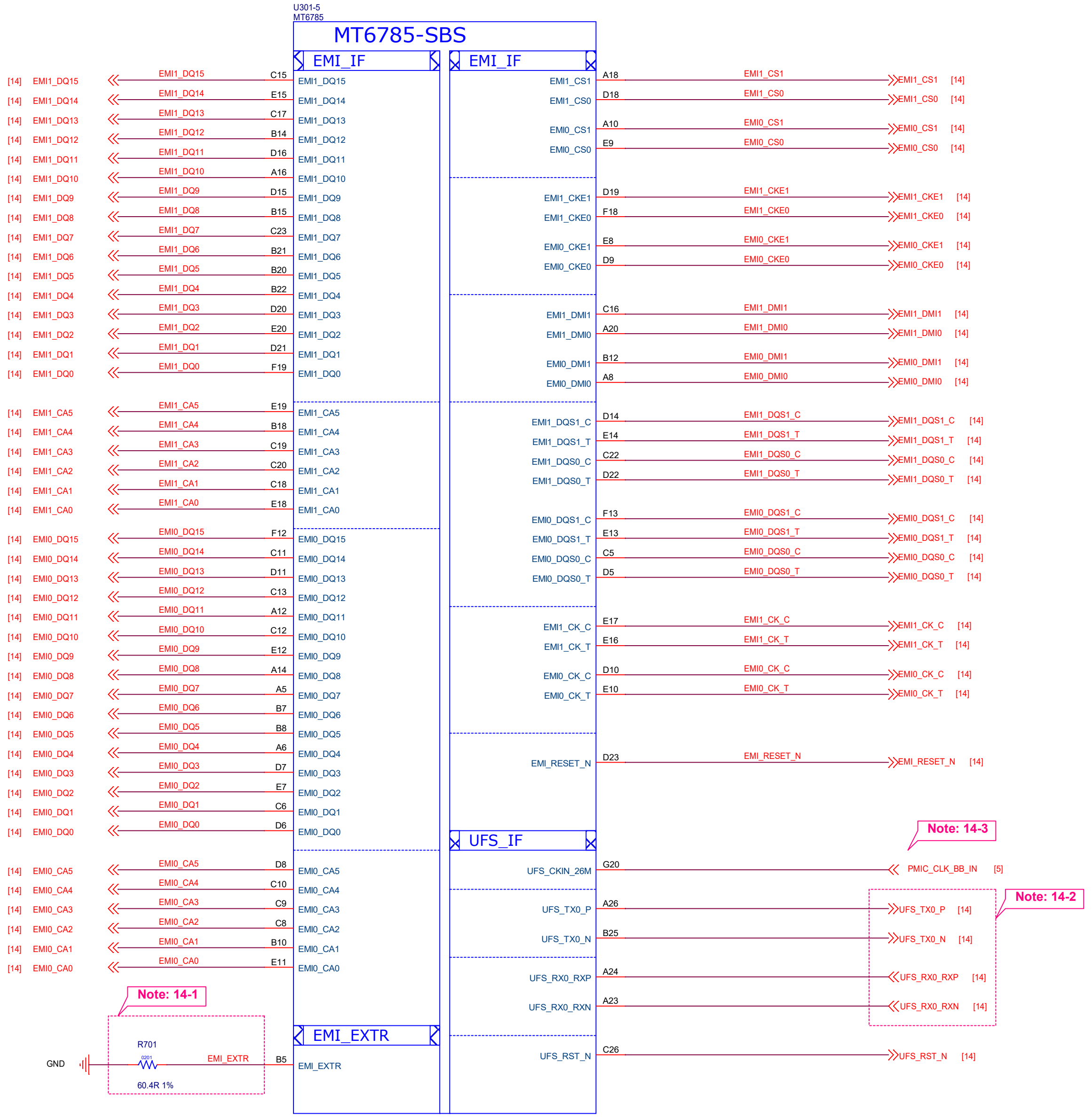

Schematic design notice of "14_BB_3_Interface" page:

Note 14-1: R4001 please select 60.4 ohm (1%) resistor

Note 14-2: Make sure TX/RX connection (T to in, R to out and P to t, N to c)

Note 14-3: UFS IP use 26MHz clock from PMIC XO_SOC (shared with ABB IF) and UFS device from PMIC XO_EXT. All of their layout must be shielded by GND.

Title	07_BB_3
Size C	MTK Confidential
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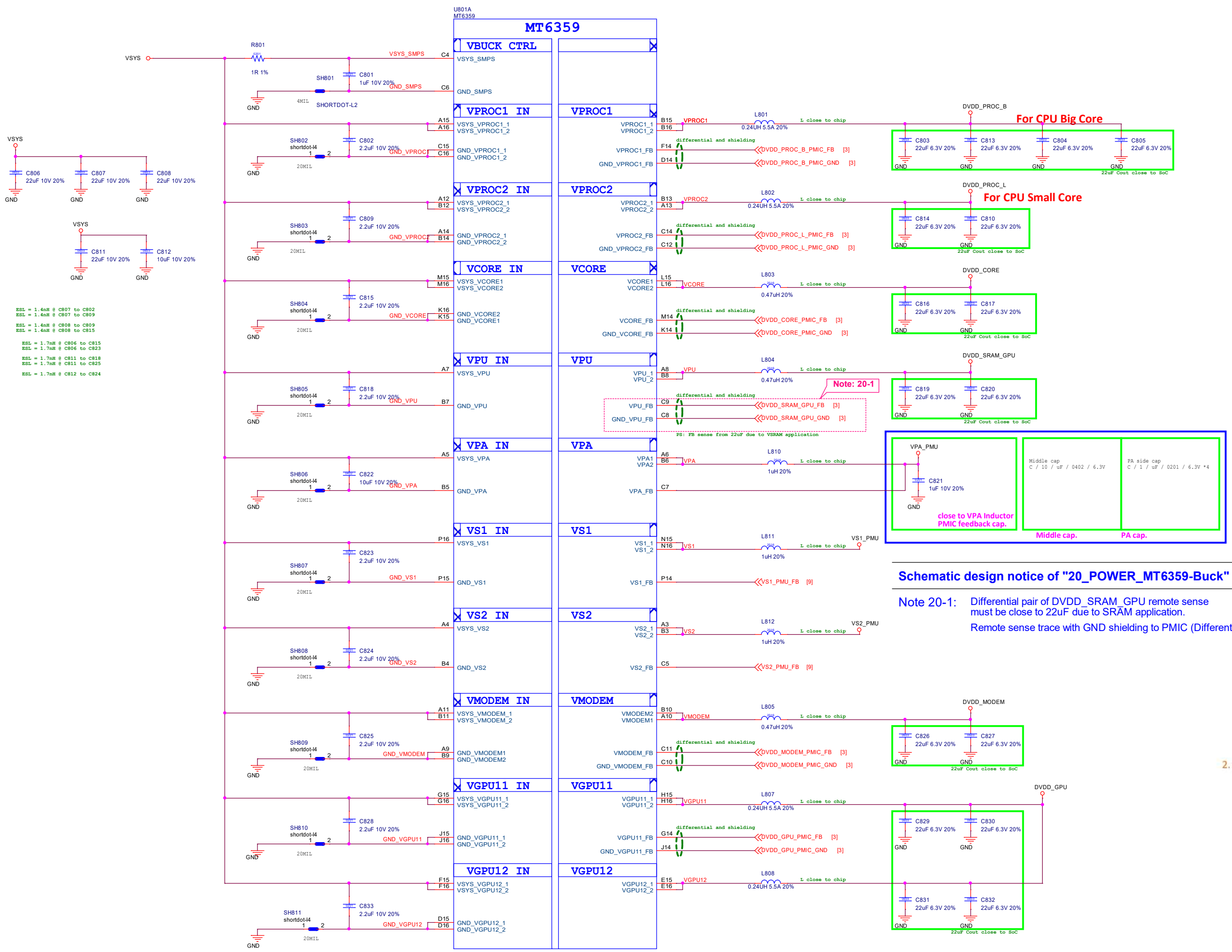

Schematic design notice of "20_POWER_MT6359-Buck" page.

Note 20-1: Differential pair of DVDD_SRAM_GPU remote sense must be close to 22uF due to SRAM application.

Remote sense trace with GND shielding to PMIC (Differential)

2. Although small size and high efficiency are major concerns, the inductor should have low core losses and low DCR (copper wire resistance). Efficiency data in MT6359 datasheet is based on 30mΩ DCR.

BUCK	Inductance (uH)	$I_{SAT,MAX}$ (A)	$I_{TEMP,MAX}$ (A)
VS1	1.0	2.6	2.0
VS2	1.0	2.6	2.0
VPA	1.0	4.0	3.4
VMODEM	0.47	4.8	4.2
VPROC1	0.24	6.3	5.0
VPROC2	0.24	6.1	4.8
VCORE	0.47	5.5	4.8
VGPU11	0.24	6.3	5.0
VGPU12	0.24	6.3	5.0
VPU (VSRAM_GPU)	0.47	1.3	0.6

MEDIATEK

Note: I_{SAT}: Depends on inductance saturation. (-30% reduction from Initial L value)

60

<Variant Name>

Title: 08_POWER_MT6359-Buck

Size: D

MTK Confidential

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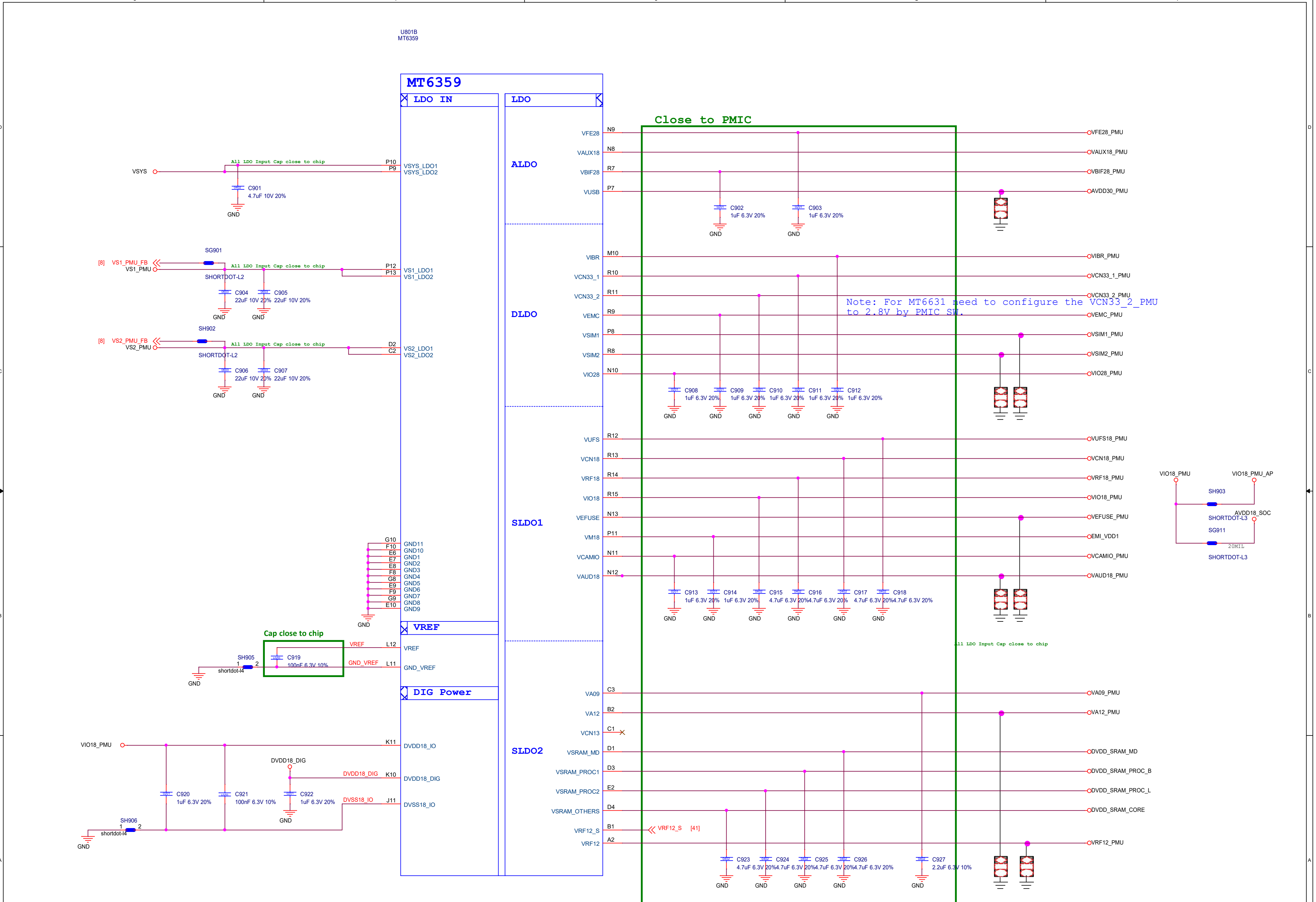
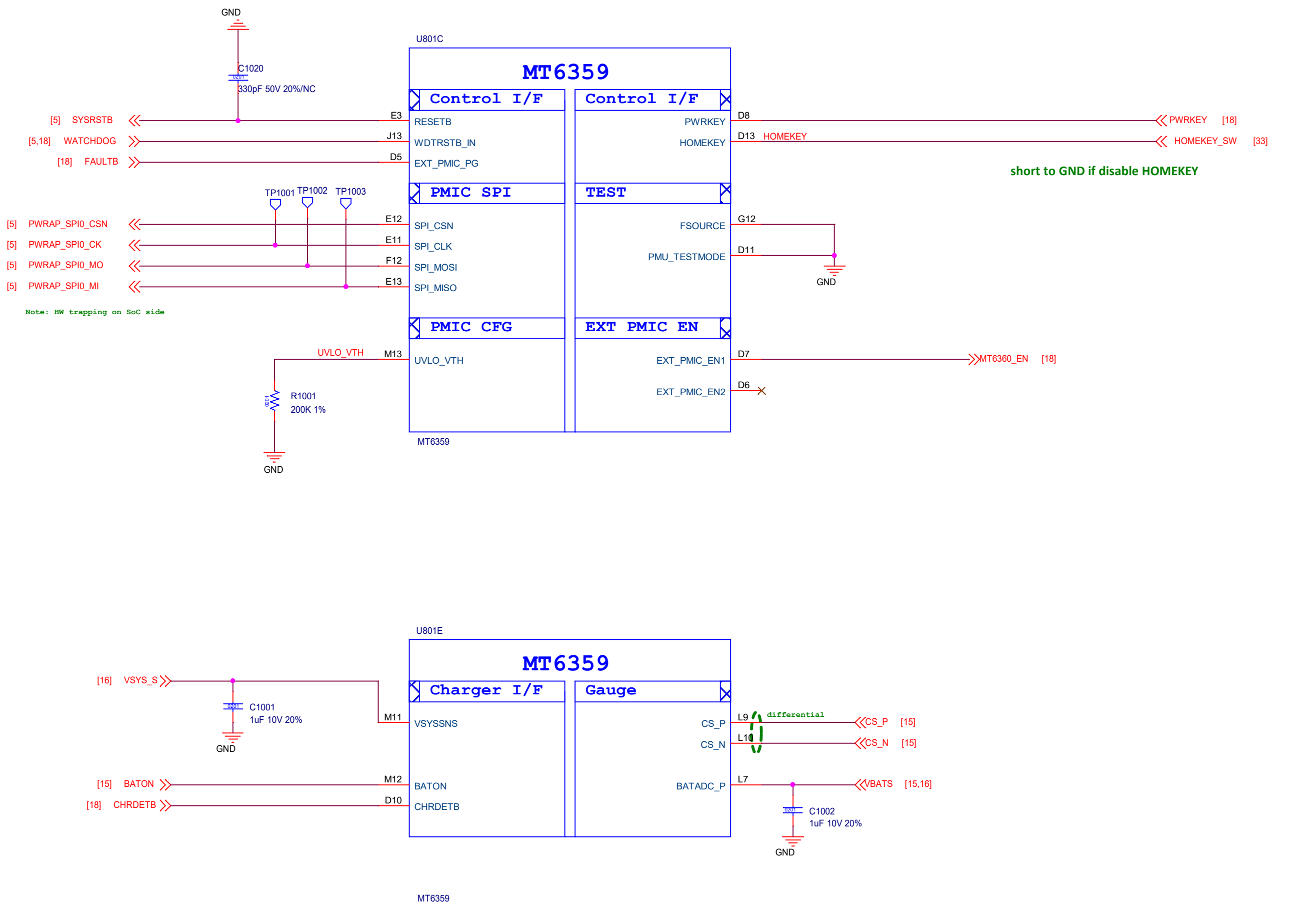


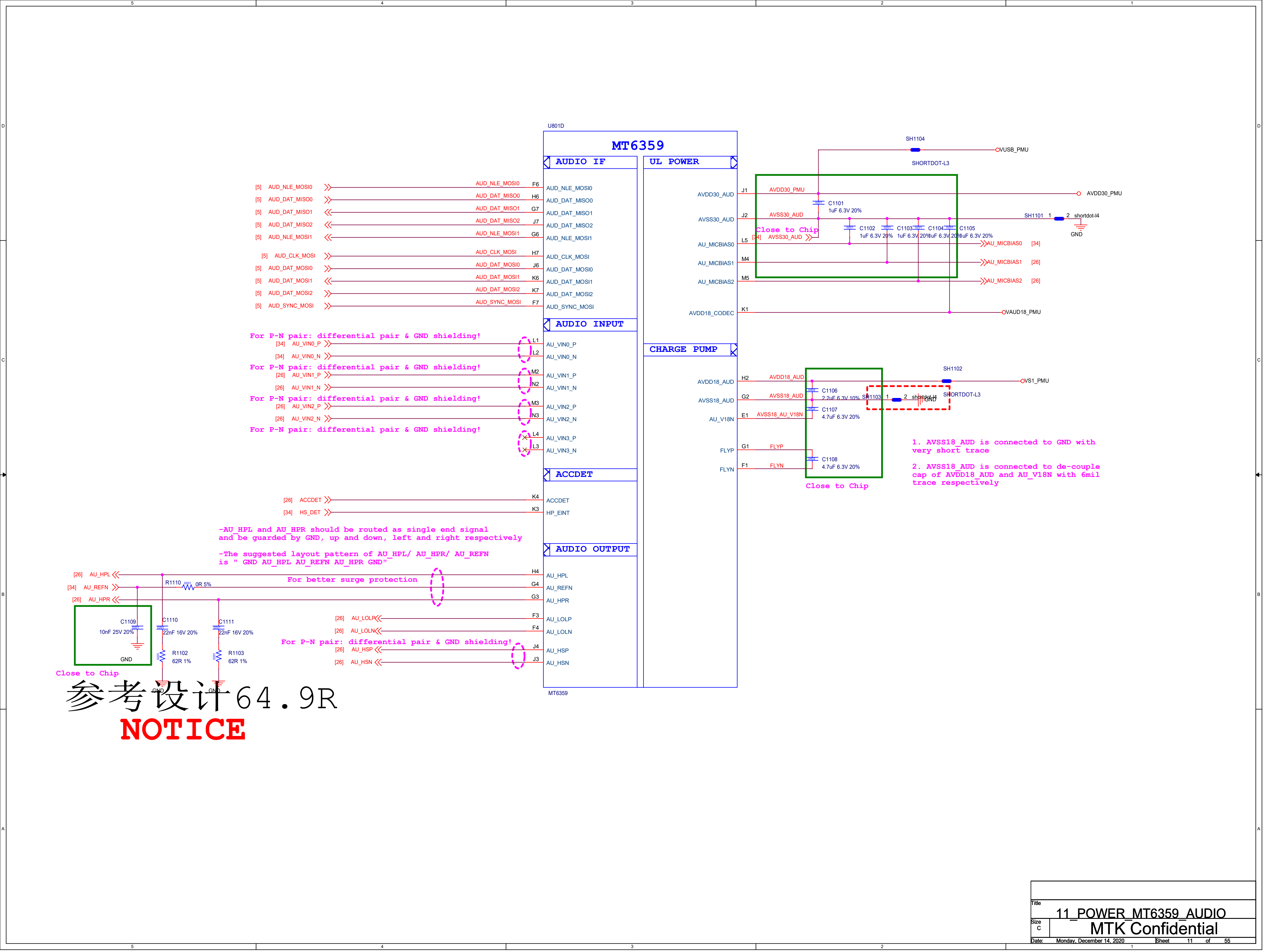
<Core Design>

Title	09_POWER_MT6359-LDO
Size C	MTK Confidential
Date: Monday, December 14, 2020	Sheet 9 of 55



<Variant Name>

Title	10_POWER_MT6359-General
Size C	MTK Confidential
Date: Monday, December 14, 2020	Sheet 10 of 55



U801D
MT6359
AUDIO IF
UL POWER
AUD_NLE_MOSI0
AUD_DAT_MISO0
AUD_DAT_MISO1
AUD_DAT_MISO2
AUD_NLE_MOSI1
AUD_CLK_MOSI
AUD_DAT_MOSI0
AUD_DAT_MOSI1
AUD_DAT_MOSI2
AUD_SYNC_MOSI
AVDD30_AUD
AVSS30_AUD
AU_MICBIAS0
AU_MICBIAS1
AU_MICBIAS2
AVDD18_CODEC
SH1104
SHORTDOT-L3
VUSB_PMU
AVDD30_PMU
C1101 1uF 6.3V 20%
SH1101 shortdot-l4
GND
Close to Chip
C1102 1uF 6.3V 20%
C1103 1uF 6.3V 20%
C1104 1uF 6.3V 20%
C1105 1uF 6.3V 20%
VAUD18_PMU
AUDIO INPUT
For P-N pair: differential pair & GND shielding!
AU_VIN0_P
AU_VIN0_N
AU_VIN1_P
AU_VIN1_N
AU_VIN2_P
AU_VIN2_N
AU_VIN3_P
AU_VIN3_N
CHARGE PUMP
AVDD18_AUD
AVSS18_AUD
AU_V18N
AVSS18_AU_V18N
FLYP
FLYN
SH1102
VS1_PMU
SH1103
SHORTDOT-L3
C1106 2.2uF 6.3V 10%
C1107 4.7uF 6.3V 20%
C1108 4.7uF 6.3V 20%
Close to Chip
1. AVSS18_AUD is connected to GND with very short trace
2. AVSS18_AUD is connected to de-couple cap of AVDD18_AUD and AU_V18N with 6mil trace respectively
ACCDET
ACCDET
HS_DET
HP_EINT
-AU_HPL and AU_HPR should be routed as single end signal and be guarded by GND, up and down, left and right respectively
-The suggested layout pattern of AU_HPL/ AU_HPR/ AU_REFN is " GND AU_HPL AU_REFN AU_HPR GND"
AUDIO OUTPUT
AU_HPL
AU_REFN
AU_HPR
R1110 0R 5%
For better surge protection
AU_LOLP
AU_LOLN
For P-N pair: differential pair & GND shielding!
AU_HSP
AU_HSN
C1109 10nF 25V 20%
C1110 22nF 16V 20%
C1111 22nF 16V 20%
R1102 62R 1%
R1103 62R 1%
GND
Close to Chip
参考设计64.9R
NOTICE
MT6359
Title
11_POWER_MT6359_AUDIO
Size C
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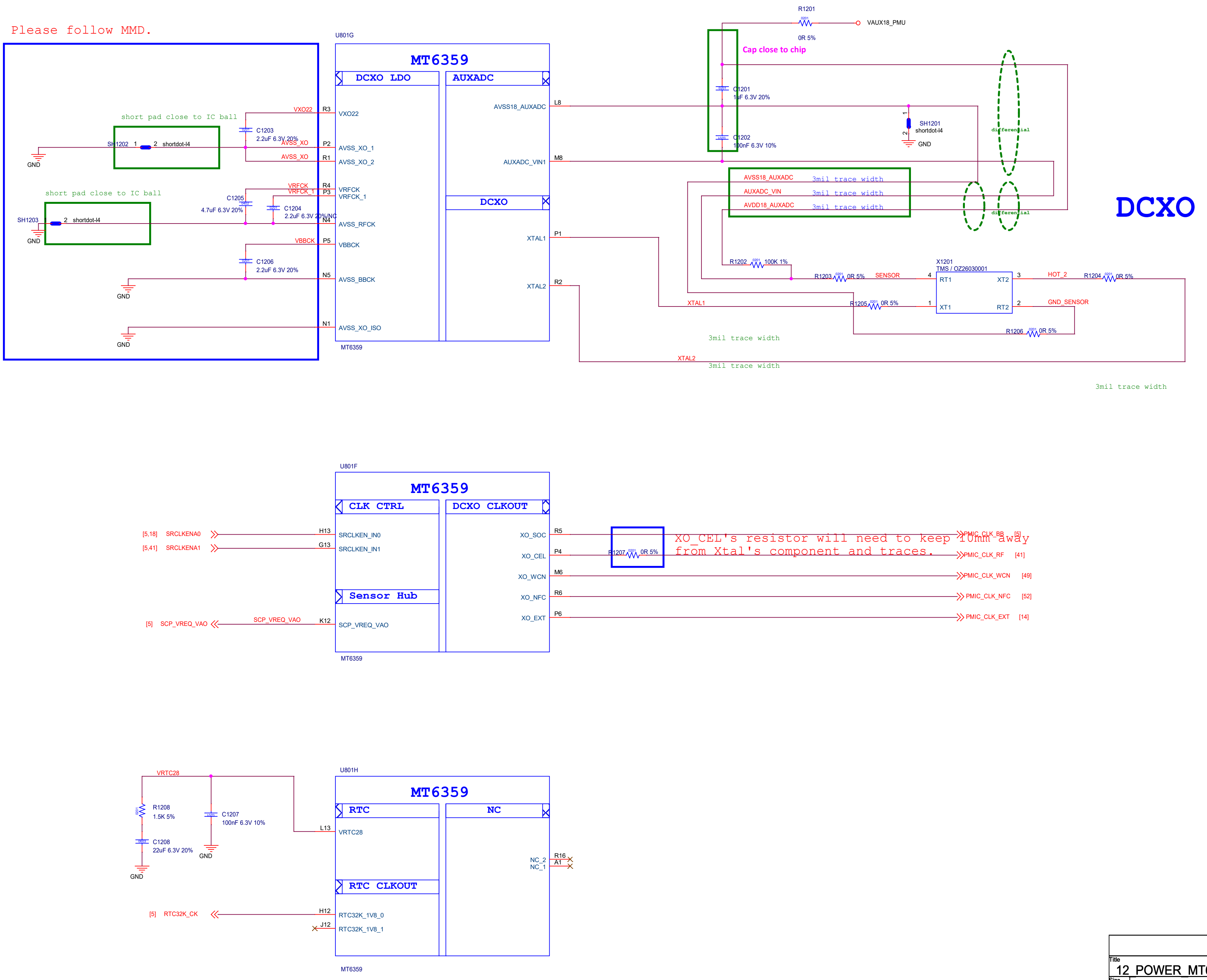

Title: 12_POWER_MT6359_Clock

Size: C

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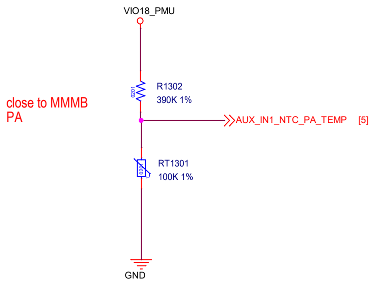
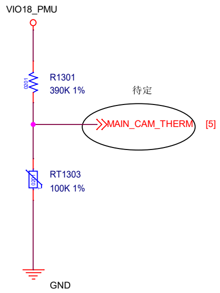
PA TEMP

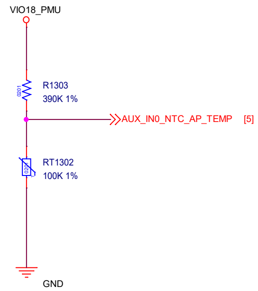
10k NTC与39k



MAIN CAM



AP TEMP



Thermistor to sense RF PA temperature

1. RT1301 must close to LTE Band 7 PA or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.

Thermistor to sense main cam temperature

1. RT1303 must close to main cam .
2. The distance is the shortest distance from package edge to edge.

Thermistor to sense AP temperature

1. RT1302 must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

Title	13_NTC_Board ID
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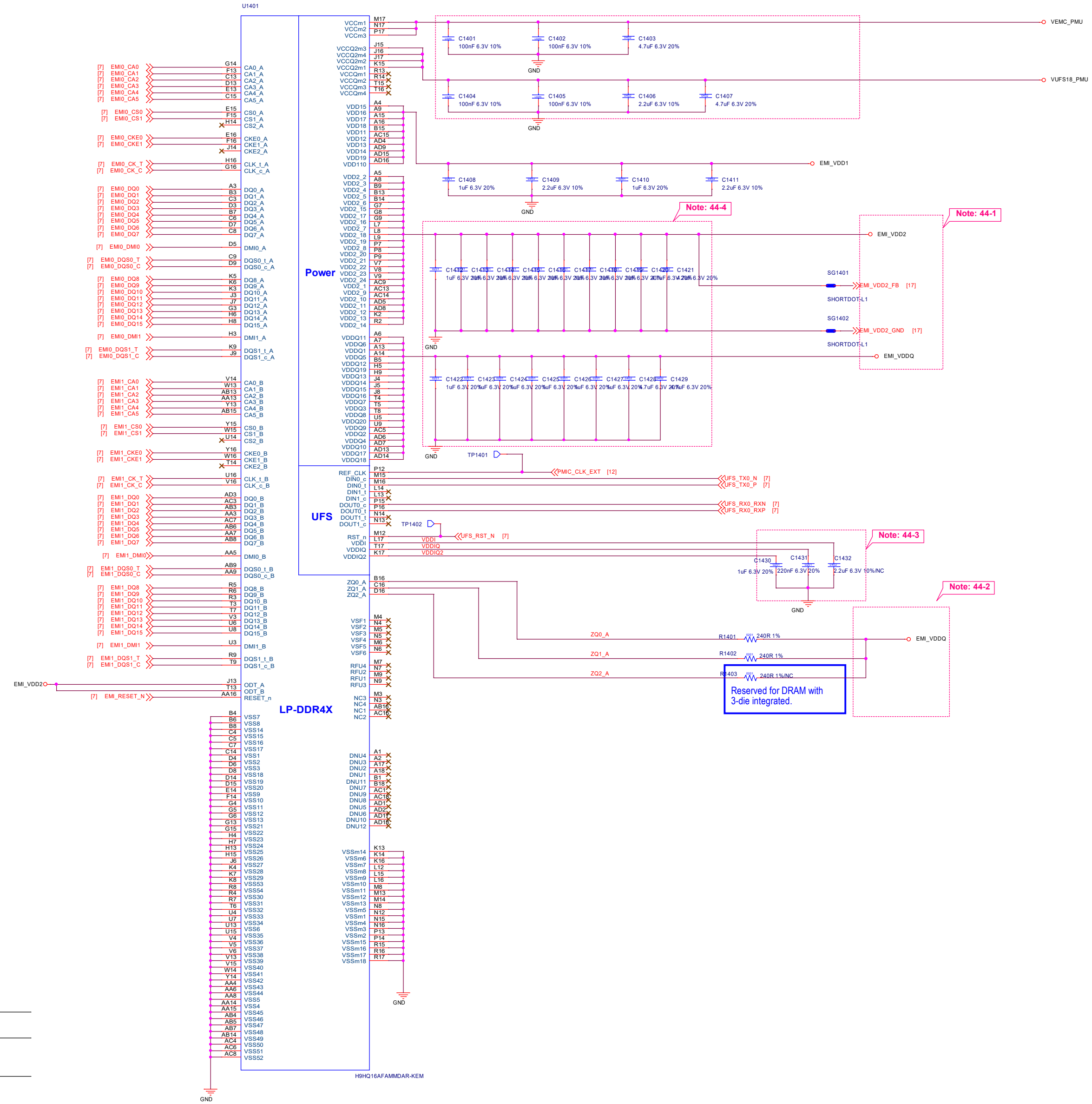

Schematic design notice of "44_Memory_UFS_LPDDR4X" page.

Note 44-1: Please refer to power supply related page select LDO7_VOUT / BUCK2_LX output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

Note 44-3: Please refer to uMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of UFS.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC [>10uF, at PMIC page]:
please also refer to MMD and layout guide for placement.

Title	14_Memory_UFS_LPDDR4X
Size D	MTK Confidential
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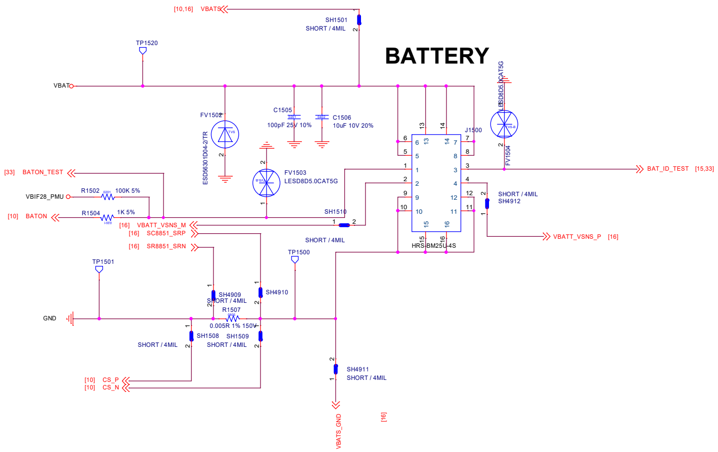
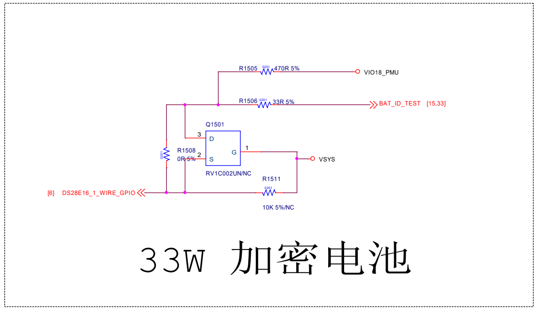
BATTERY





Title
15_Battery Fuel guage
Size D
MTK Confidential
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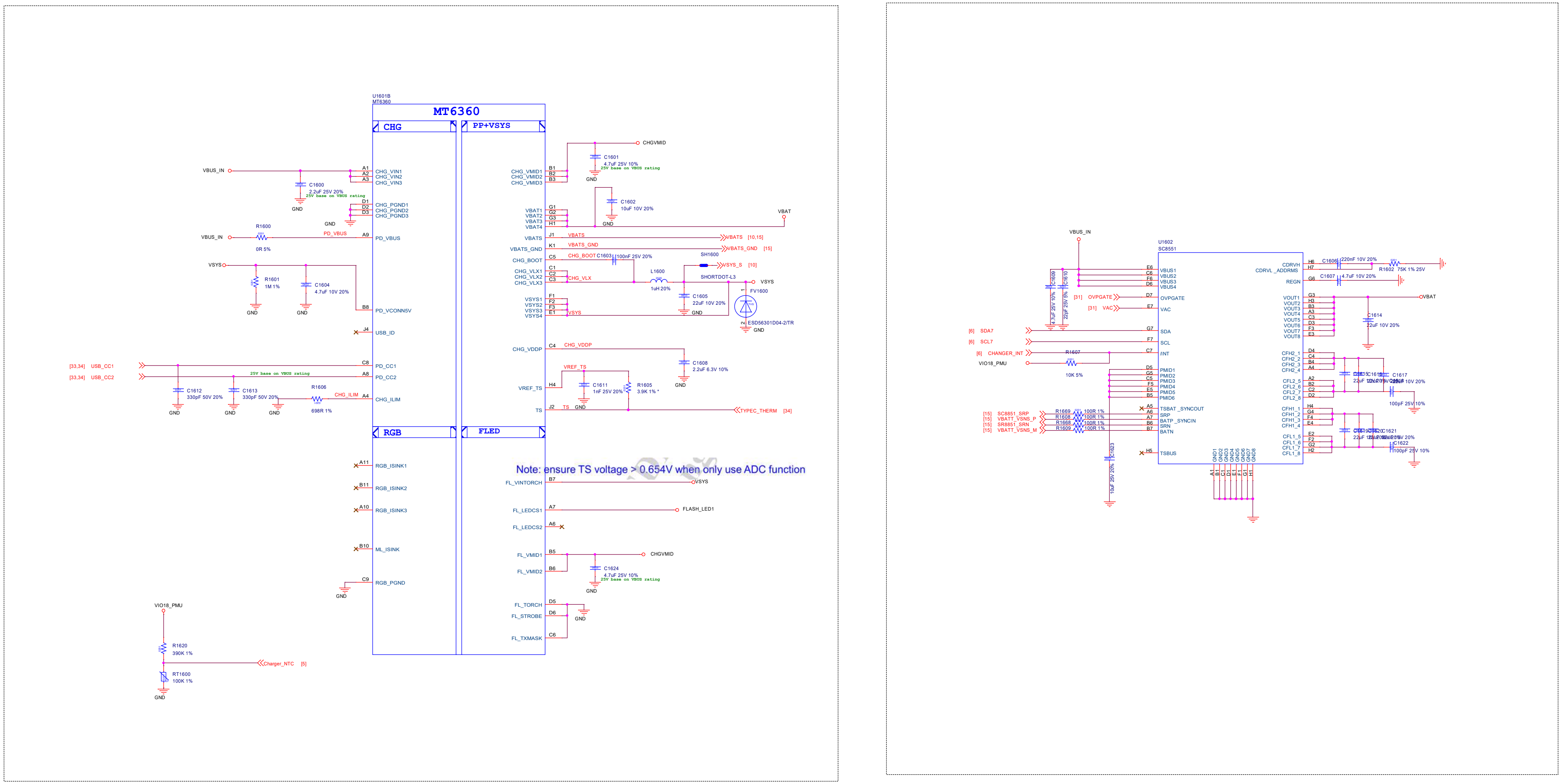
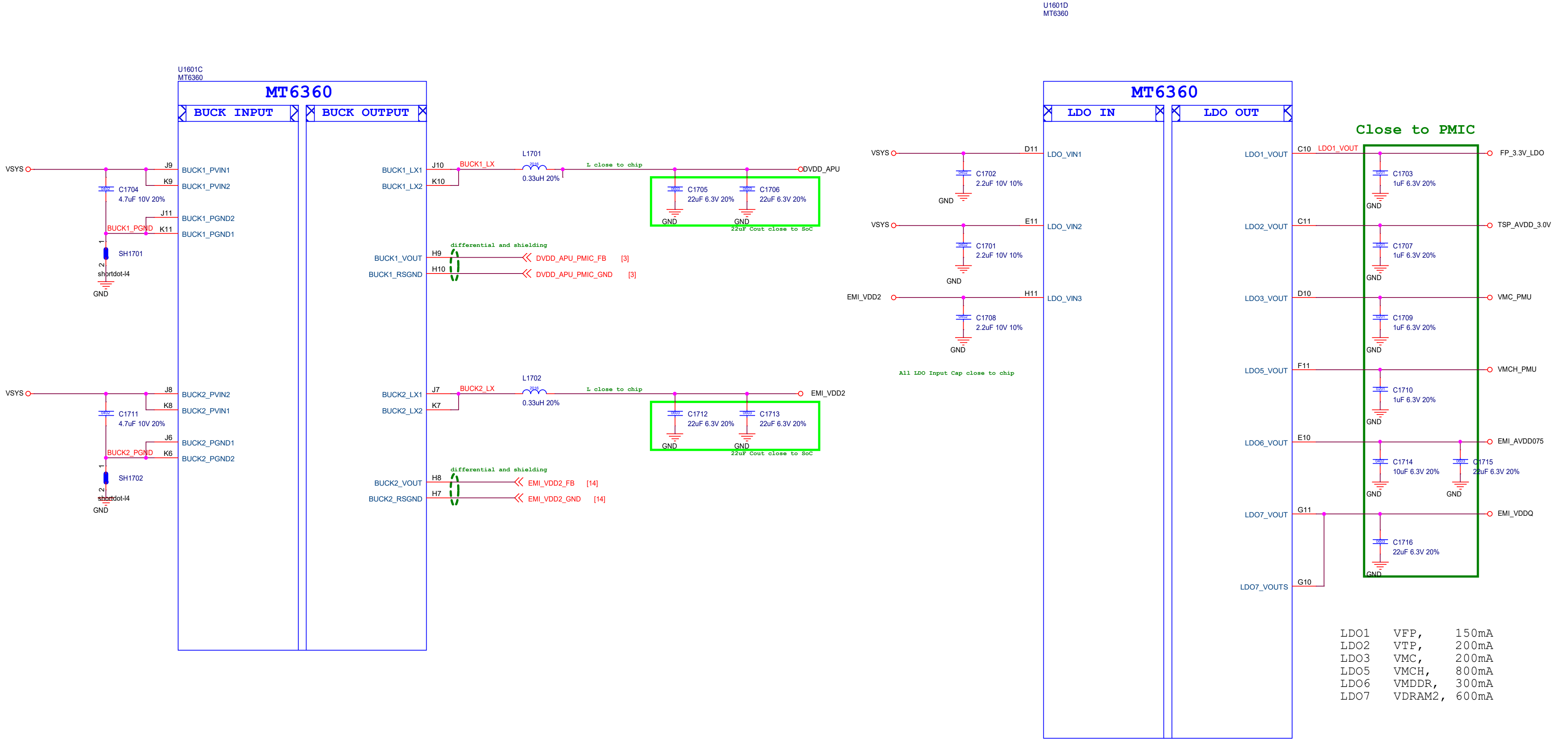


Title
16_POWER_MT6360_Charger
Size D
MTK Confidential
Date: Monday, December 14, 2020 Sheet 16 of 55



Title
17_POWER_MT6360_BUCK_LDO

Size C

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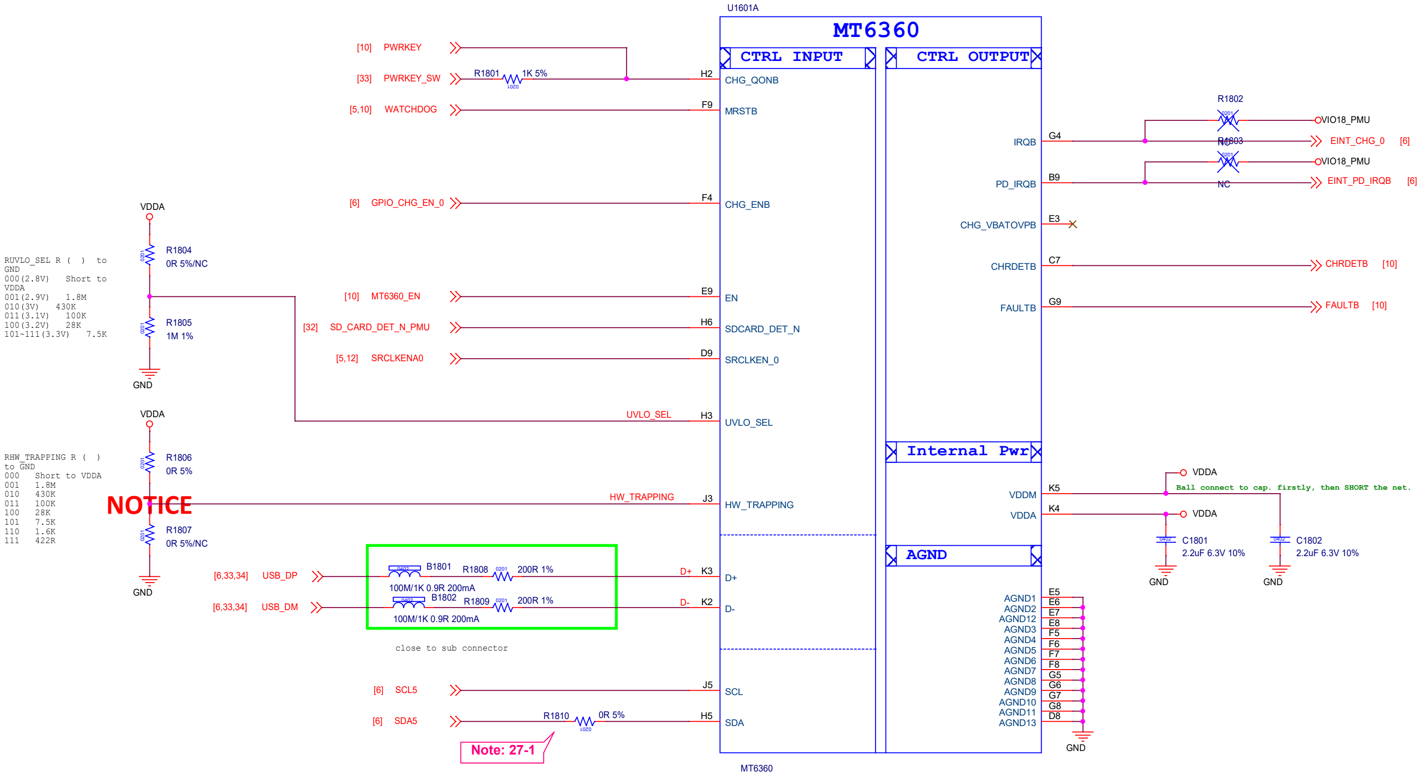

Schematic design notice of "27_POWER_MT6360_General" page:

Note 27-1: To optimize driving and PCB routing , it is recommended to reserve R for BOM fine tuning.
The R vaule and rule refer to MT6360 desing notice.

Title	18_POWER_MT6360_General
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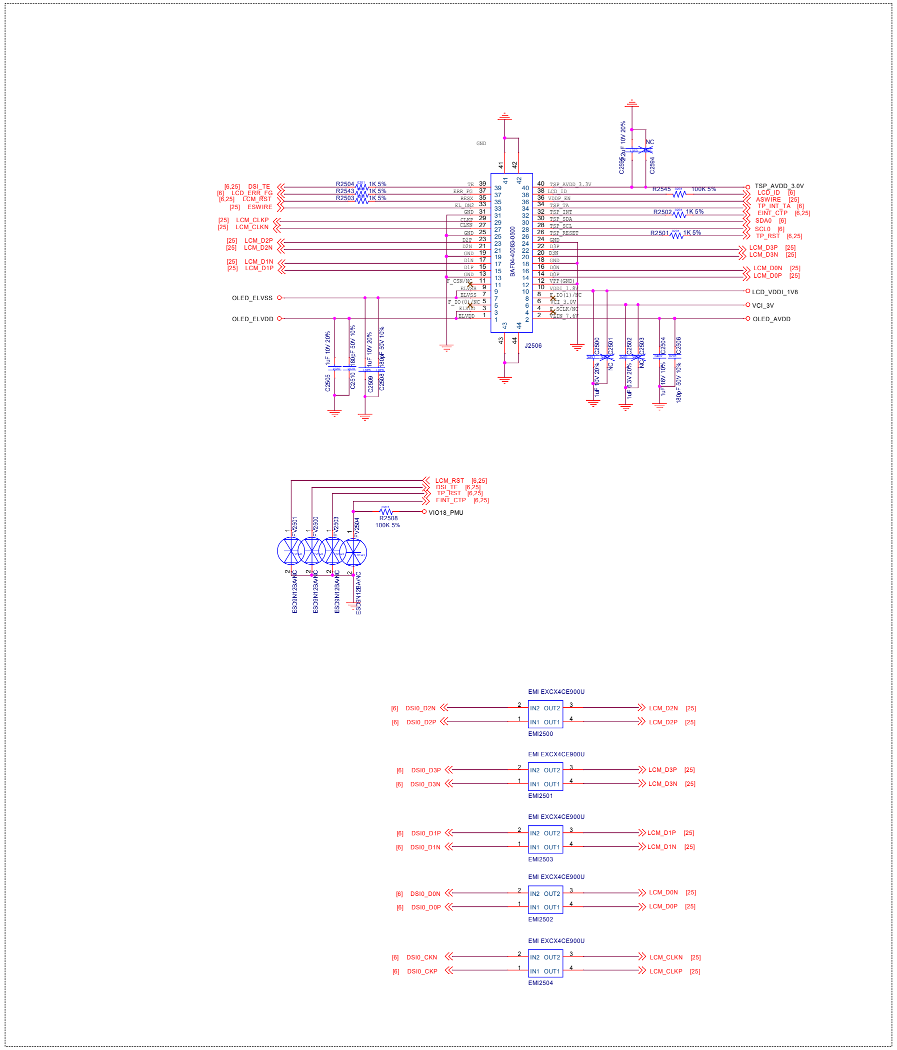

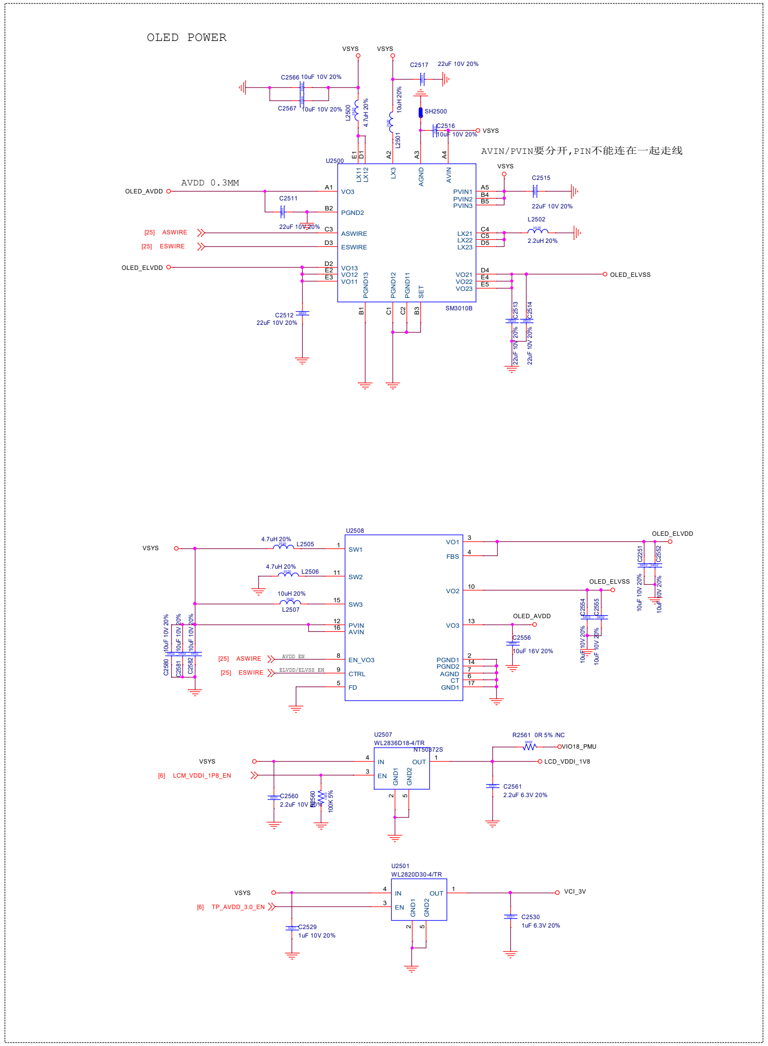

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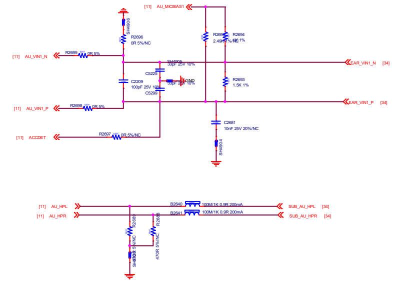
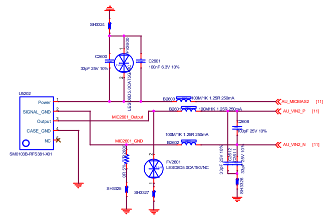
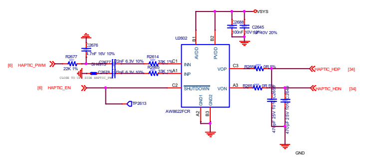
Earphone Audio



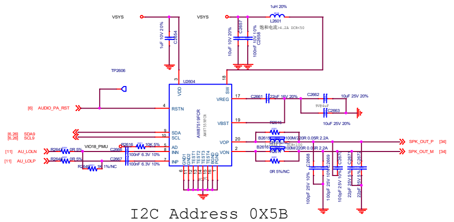
Sub MIC



HAPTIC

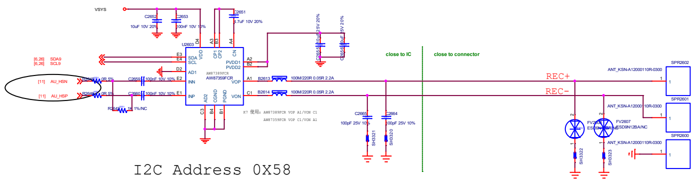


PA FOR SPK

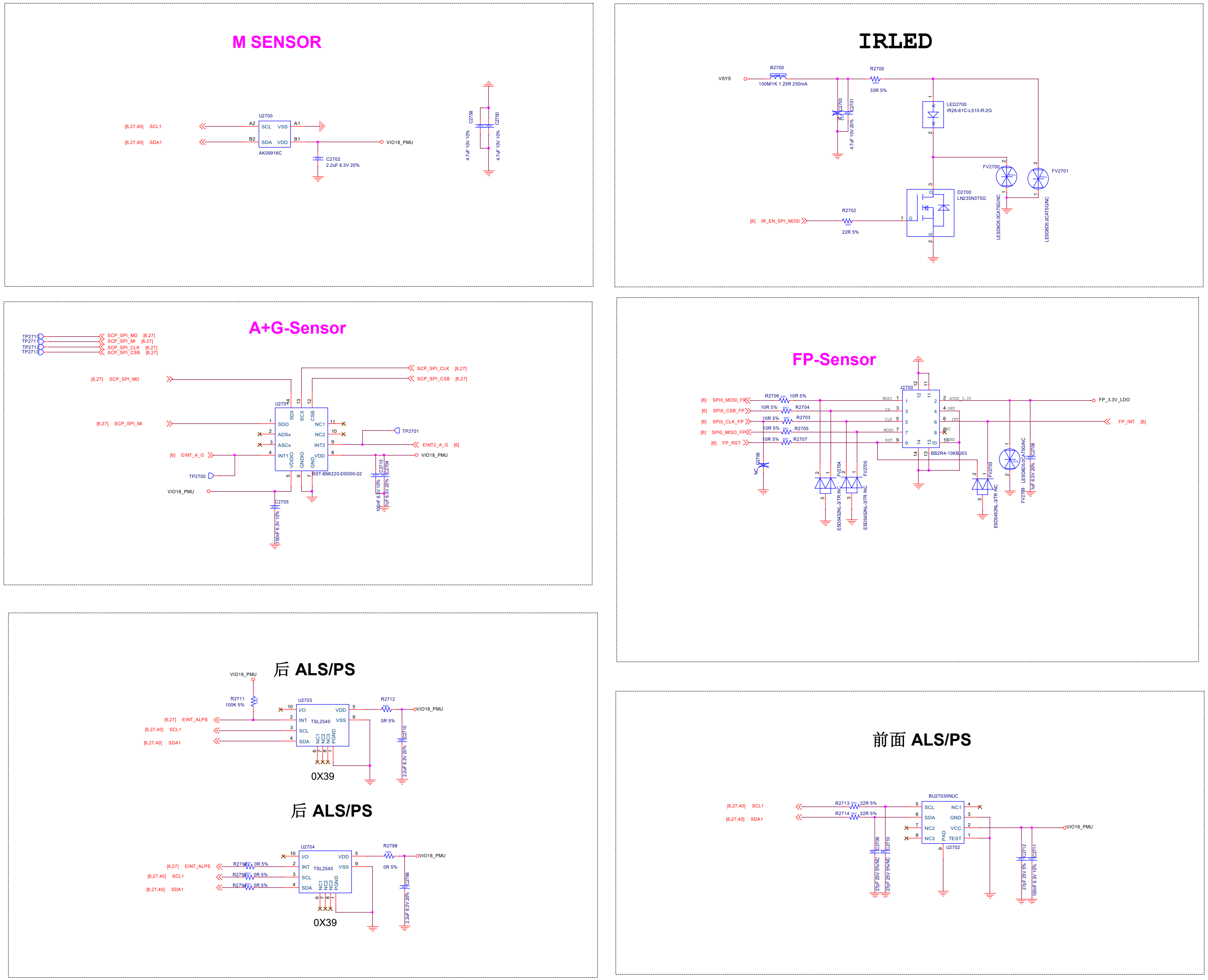


I2C Address 0X5B

PA FOR REC



I2C Address 0X58



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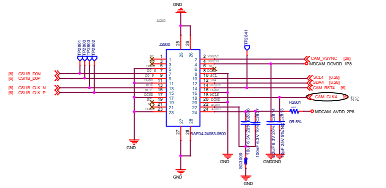
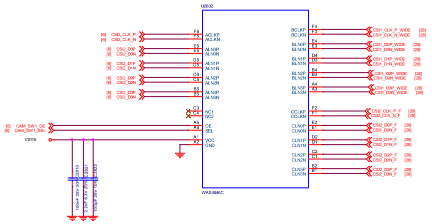
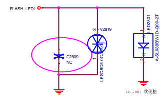
Depth Camera 2M



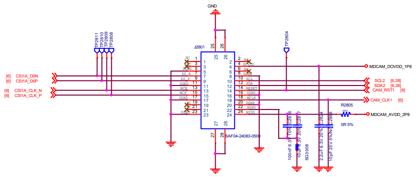
MIPI SW1



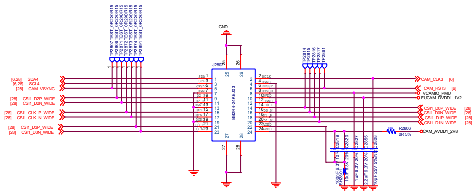
FLASH LED



Micro Camera 5M

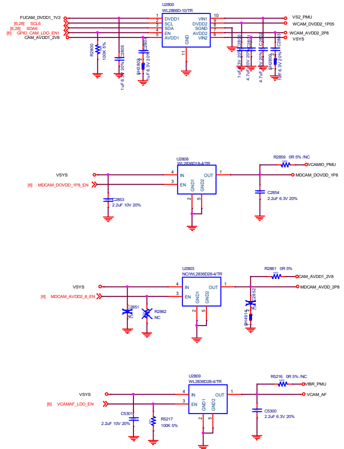


WIDE Camera 8M



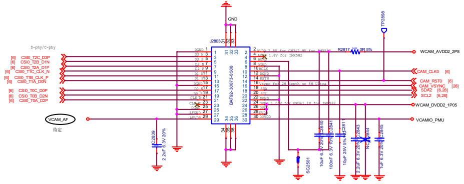
POWER LDO

I2C Address 0X40

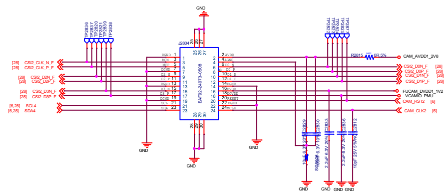


	AVDD	DVDD	IOVDD	AFVDD	I2C	MCLK	PDN	RESET
F_CAM	CAM_AVDD1_2V8	FUCAM_DVDD1_1V2	VCAMIO_PMU		4	MCLK2		CAM2_RST_N
B_CAM	WCAM_AVDD2_2P8	WCAM_DVDD2_1P05	VCAMIO_PMU	VCAM_AF/VIBR_PMU	2	MCLK0		CAM0_RST_N
M_CAM	CAM_AVDD1_2V8		MDCAM_DOVDD_1P8		2	MCLK1		CAM1_RST_N
D_CAM	CAM_AVDD1_2V8		MDCAM_DOVDD_1P8		4	MCLK4		CAM4_RST_N
W_CAM	CAM_AVDD1_2V8	FUCAM_DVDD1_1V2	VCAMIO_PMU		6	MCLK3		CAM3_RST_N

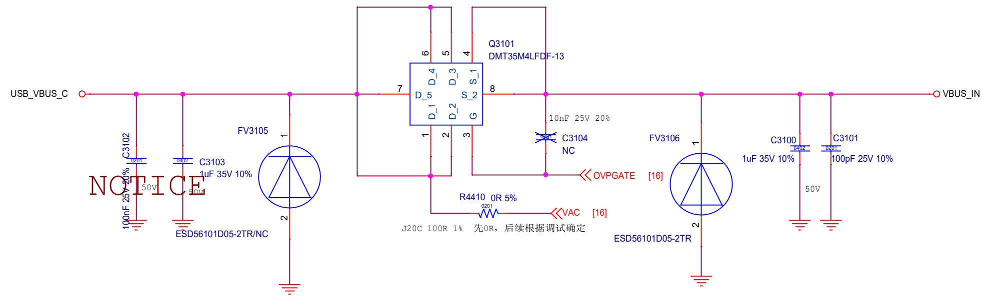
Main Camera 64M



Front Camera 13M/16M



OVP



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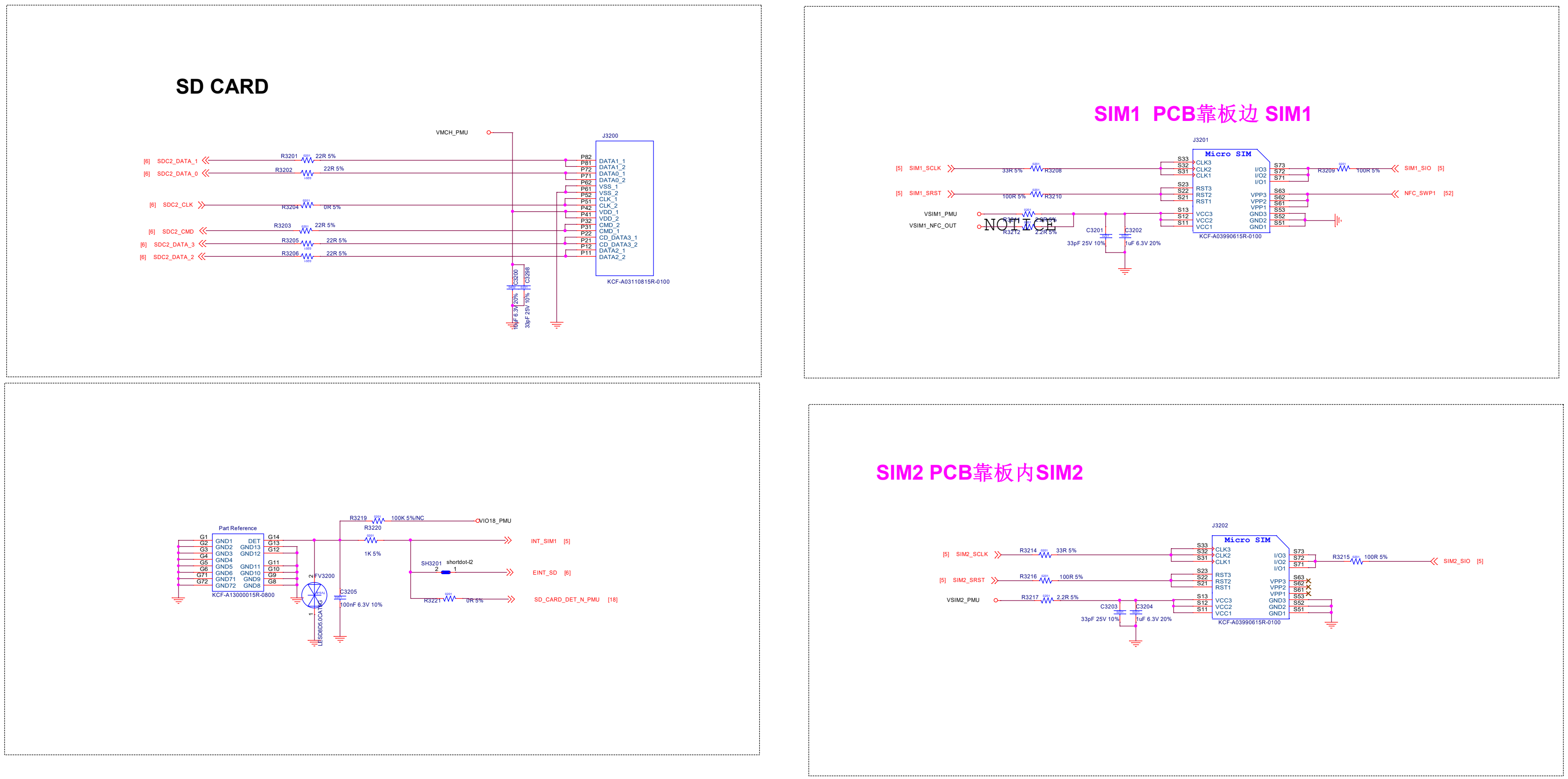

Title
<Title>

Size D

Document Number
<Doc>

Rev
<RevCode>

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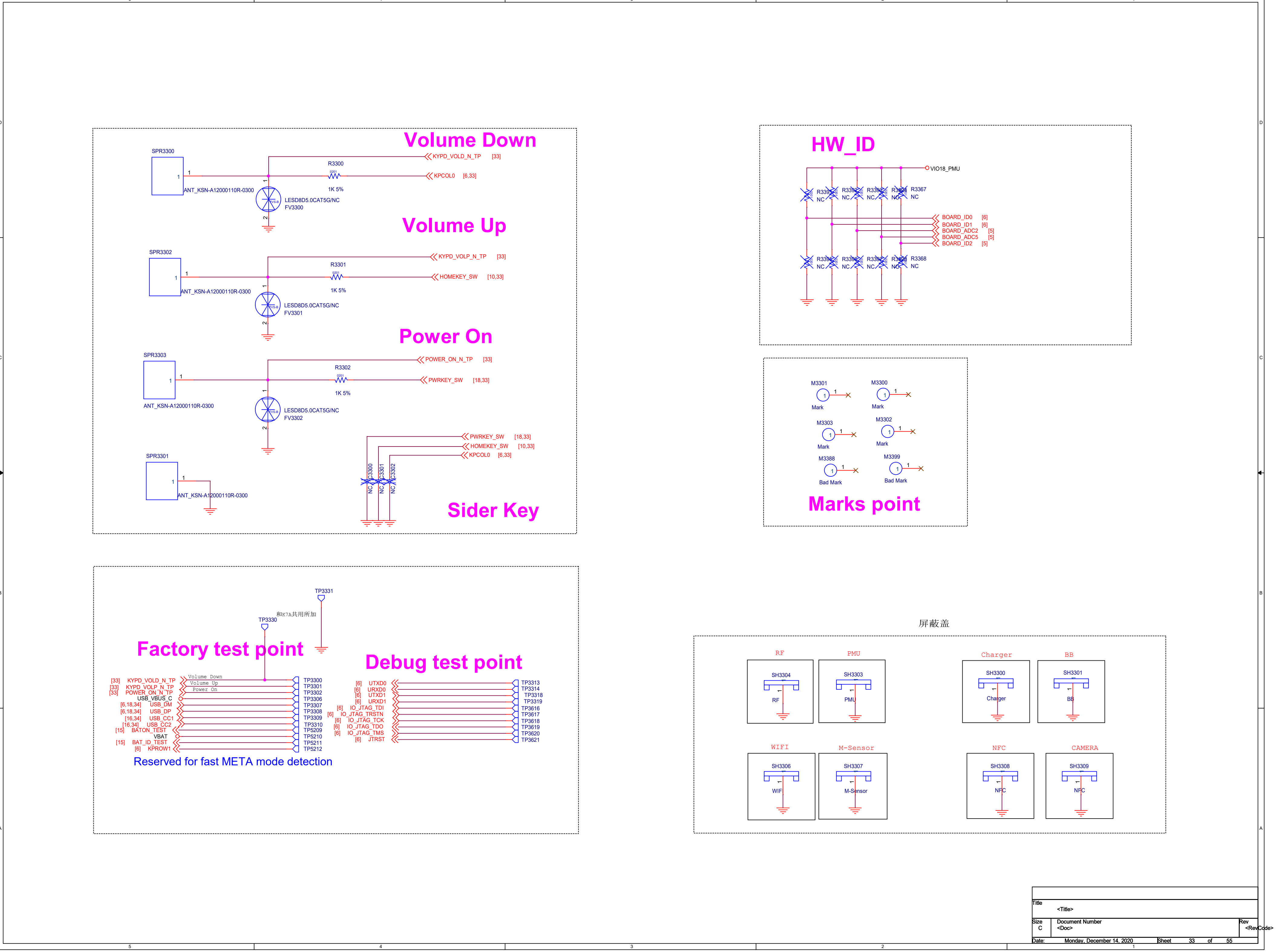

Volume Down
SPR3300
ANT_KSN-A12000110R-0300
R3300
1K 5%
KYPD_VOLD_N_TP [33]
KPCOL0 [6,33]
LESD8D5.0CAT5G/NC
FV3300
Volume Up
SPR3302
ANT_KSN-A12000110R-0300
R3301
1K 5%
KYPD_VOLP_N_TP [33]
HOMEKEY_SW [10,33]
LESD8D5.0CAT5G/NC
FV3301
Power On
SPR3303
ANT_KSN-A12000110R-0300
R3302
1K 5%
POWER_ON_N_TP [33]
PWRKEY_SW [18,33]
LESD8D5.0CAT5G/NC
FV3302
SPR3301
ANT_KSN-A12000110R-0300
PWRKEY_SW [18,33]
HOMEKEY_SW [10,33]
KPCOL0 [6,33]
NC C3300
NC C3301
NC C3302
Sider Key
HW_ID
VIO18_PMU
R3367 NC
R3368 NC
BOARD_ID0 [6]
BOARD_ID1 [6]
BOARD_ADC2 [5]
BOARD_ADC5 [5]
BOARD_ID2 [5]
M3301 Mark
M3300 Mark
M3303 Mark
M3302 Mark
M3388 Bad Mark
M3399 Bad Mark
Marks point
TP3331
TP3330
和K7A共用所加
Factory test point
[33] KYPD_VOLD_N_TP
[33] KYPD_VOLP_N_TP
[33] POWER_ON_N_TP
USB_VBUS_C
[6,18,34] USB_DM
[6,18,34] USB_DP
[16,34] USB_CC1
[16,34] USB_CC2
[15] BATON_TEST
VBAT
[15] BAT_ID_TEST
[6] KPROW1
Volume Down
Volume Up
Power On
TP3300
TP3301
TP3302
TP3306
TP3307
TP3308
TP3309
TP3310
TP5209
TP5210
TP5211
TP5212
Reserved for fast META mode detection
Debug test point
[6] UTXD0
[6] URXD0
[6] UTXD1
[6] URXD1
[6] IO_JTAG_TDI
[6] IO_JTAG_TRSTN
[6] IO_JTAG_TCK
[6] IO_JTAG_TDO
[6] IO_JTAG_TMS
[6] JTRST
TP3313
TP3314
TP3318
TP3319
TP3616
TP3617
TP3618
TP3619
TP3620
TP3621
屏蔽盖
RF
SH3304
PMU
SH3303
Charger
SH3300
BB
SH3301
WIFI
SH3306
M-Sensor
SH3307
NFC
SH3308
CAMERA
SH3309
Title
<Title>
Size C
Document Number
<Doc>
Rev
<RevCode>
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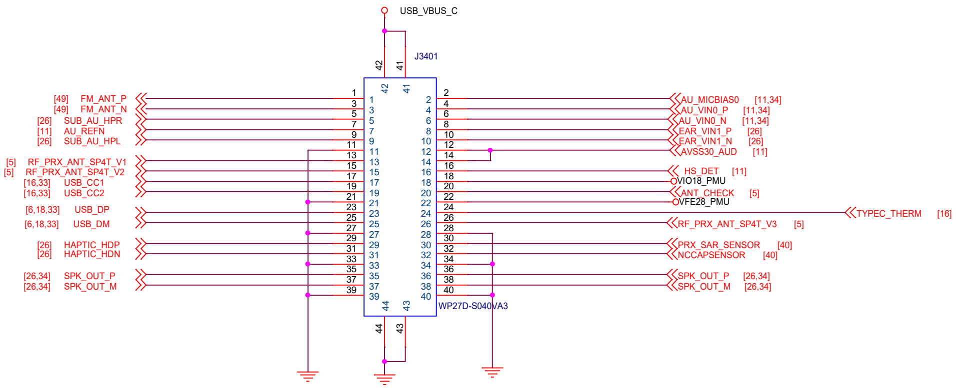
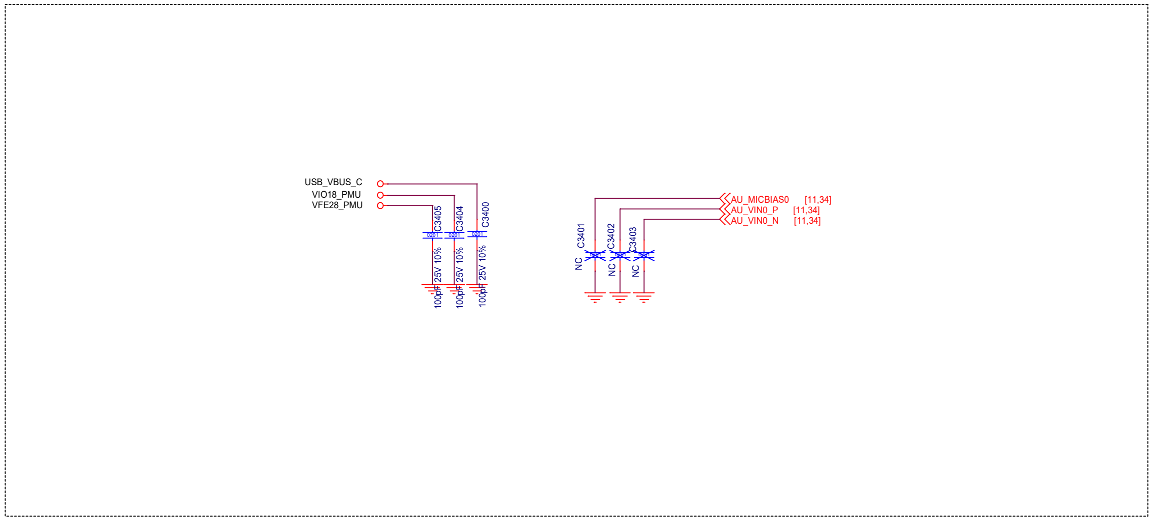


主副板连接器 1PIN位置还需要确定



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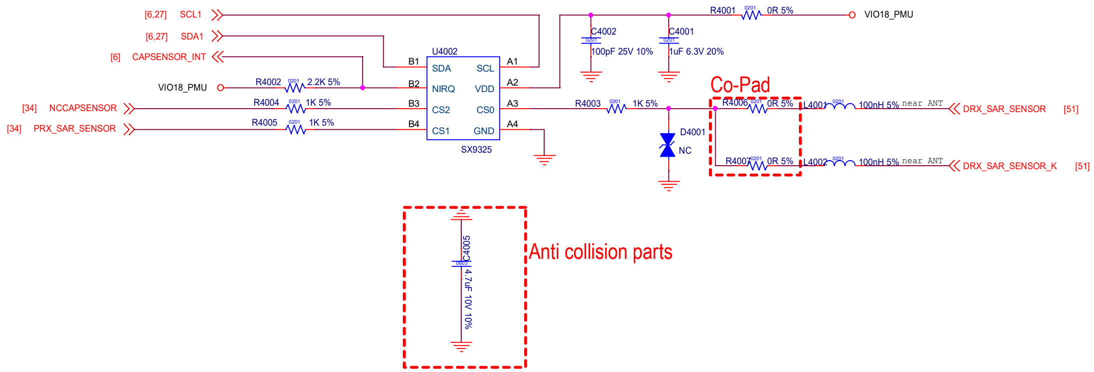
40_SAR_Sensor



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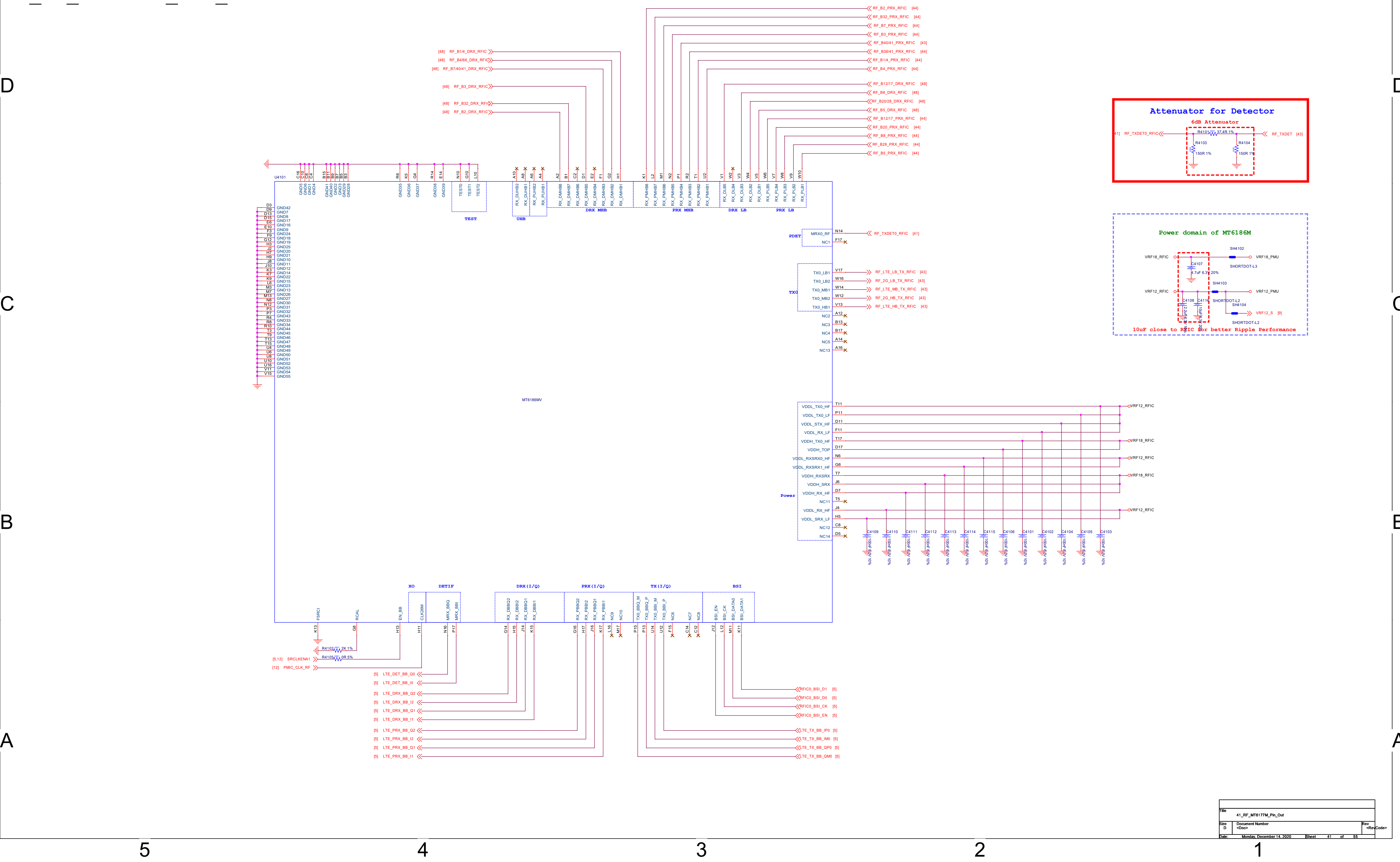
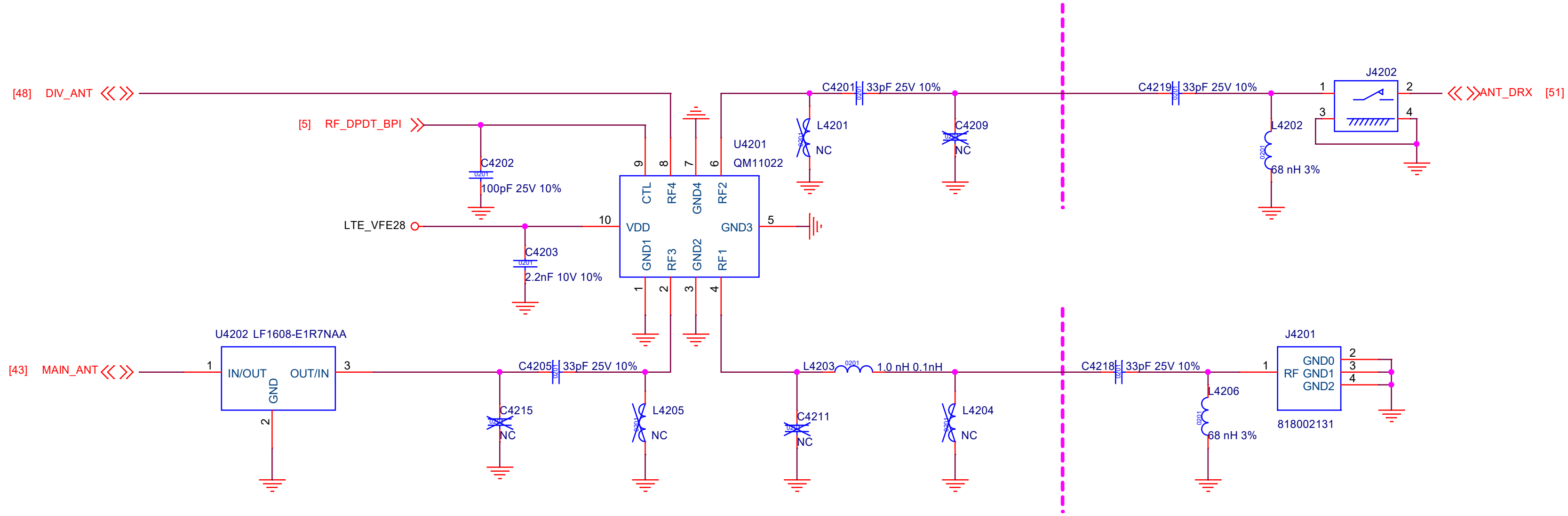
41_RF_MT6186M_Pin_Out



42_DPDT_Connector



Title	42_DPDT_Connector		
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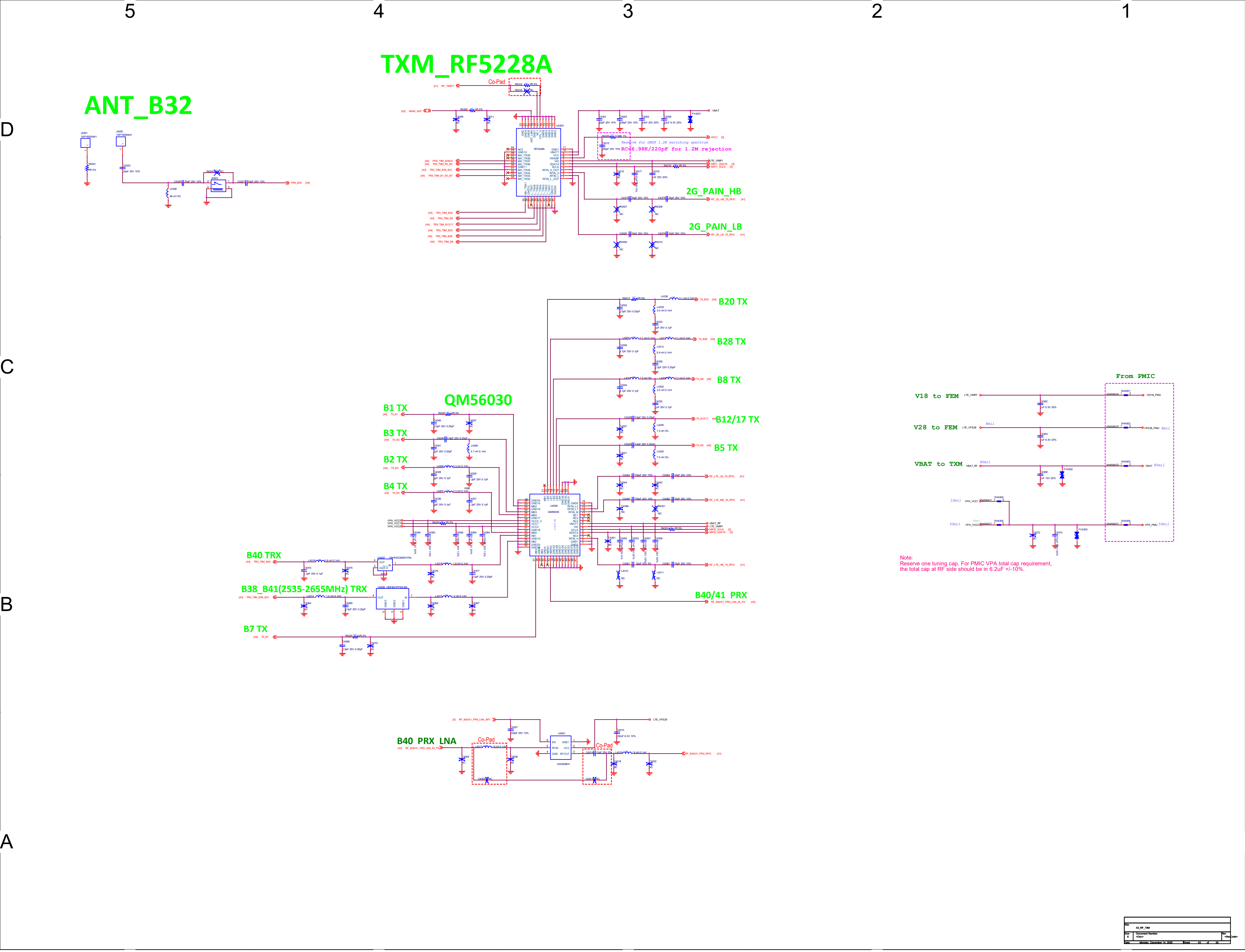
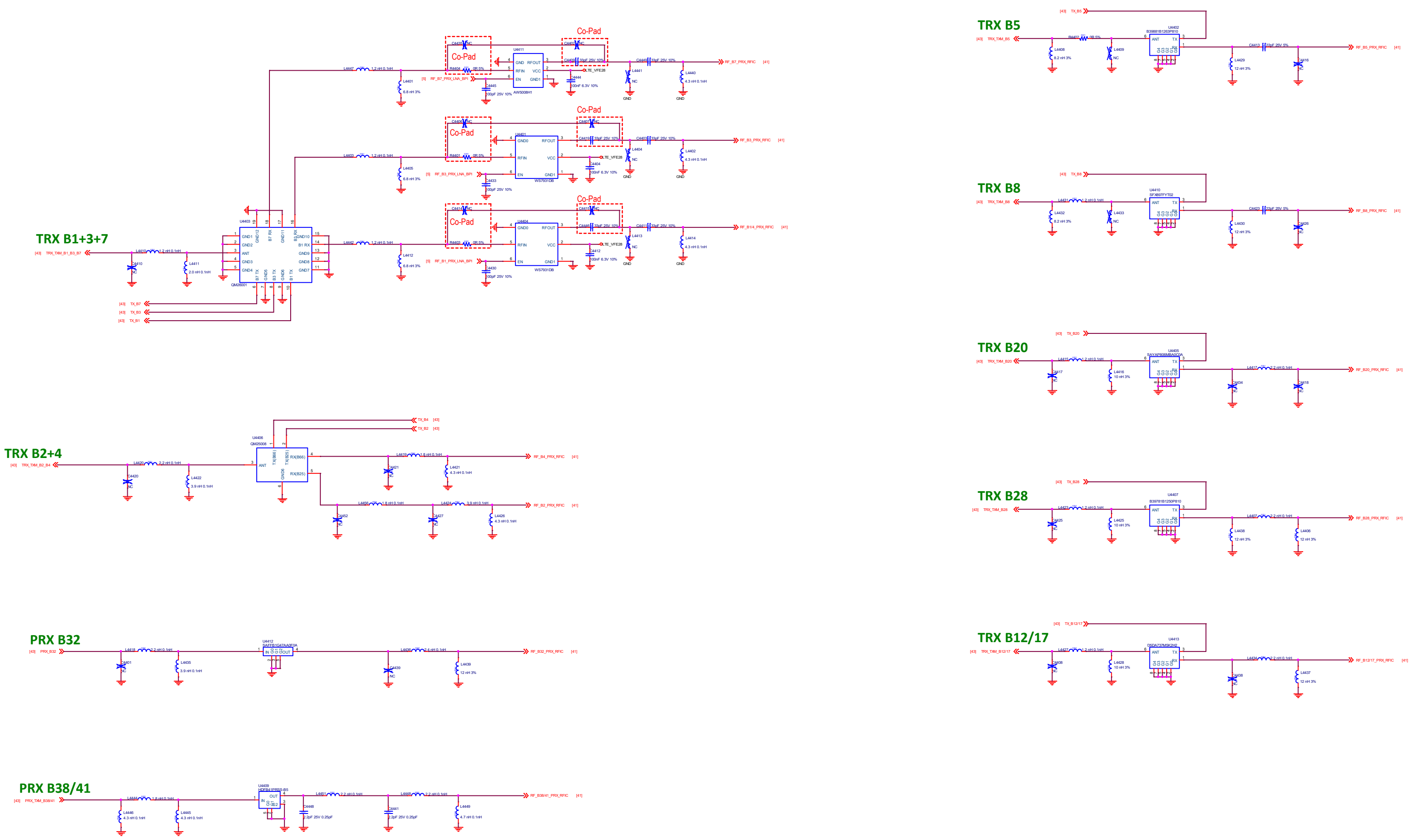
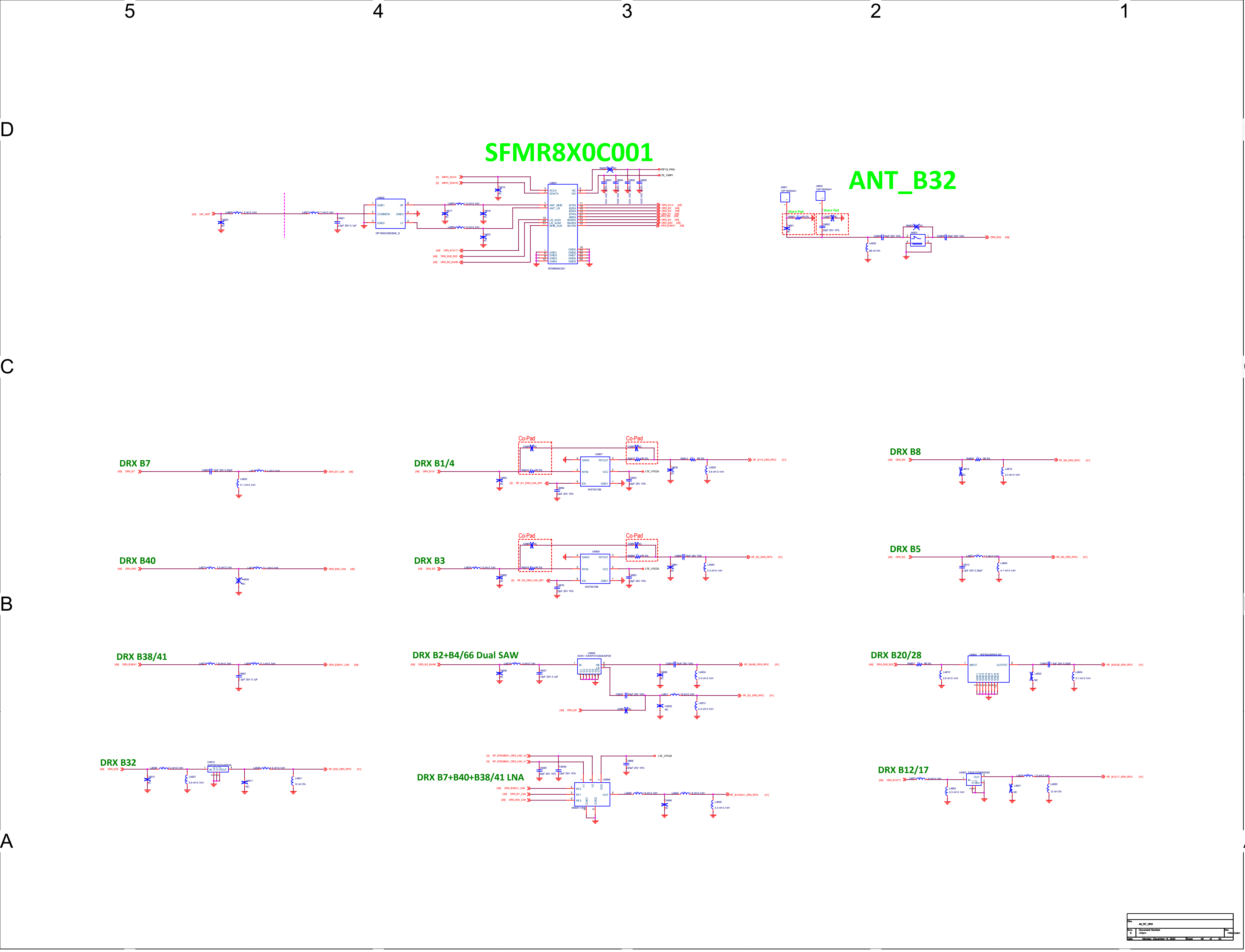
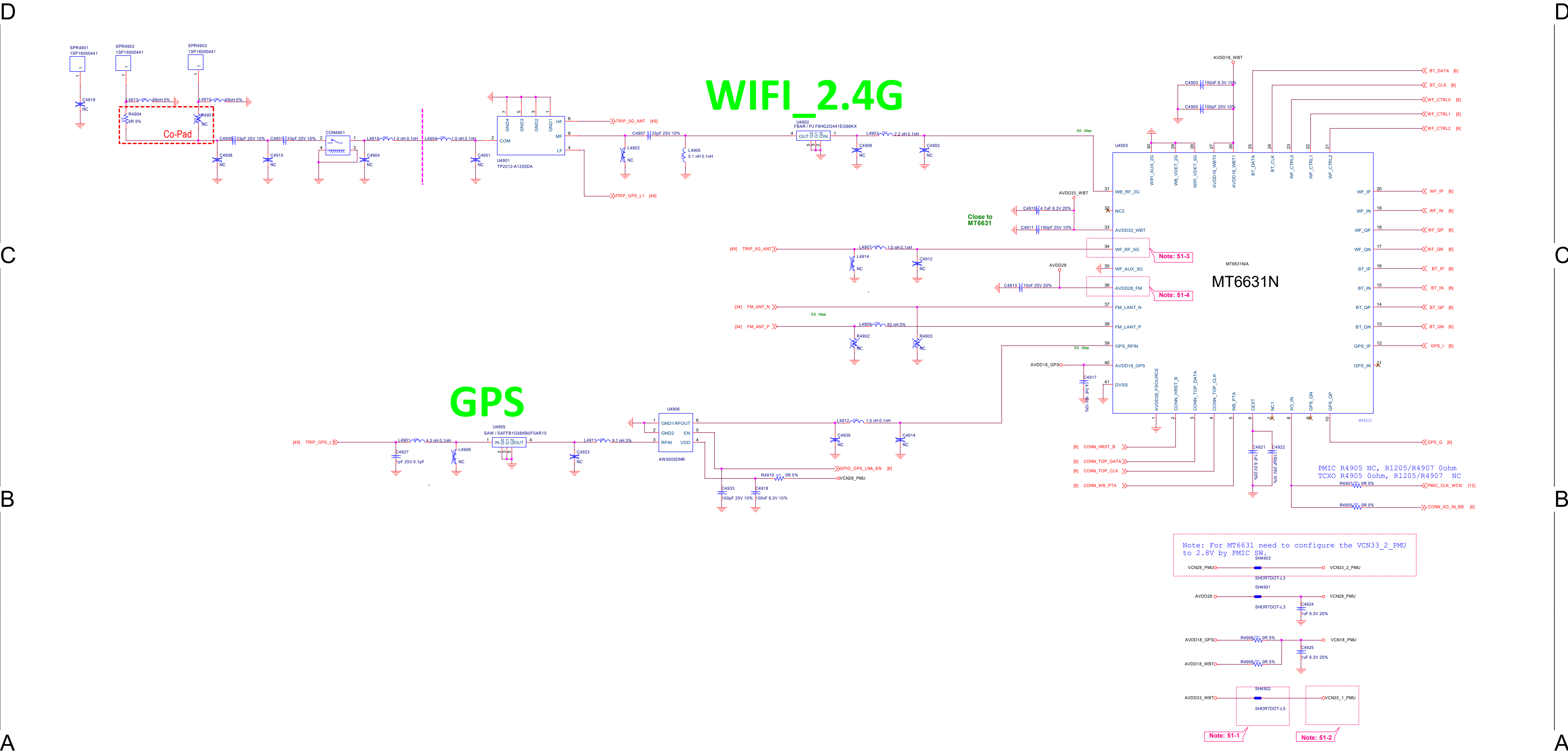

ANT_B32
TXM_RF5228A
Co-Pad
2G_PAIN_HB
2G_PAIN_LB
RC46.98K/220pF for 1.2M rejection
QM56030
B20 TX
B28 TX
B8 TX
B12/17 TX
B5 TX
B1 TX
B3 TX
B2 TX
B4 TX
B40 TRX
B38_B41(2535-2655MHz) TRX
B7 TX
B40/41 PRX
B40 PRX LNA
From PMIC
V18 to FEM
V28 to FEM
VBAT to TXM
Note:
Reserve one tuning cap. For PMIC VPA total cap requirement,
the total cap at RF side should be in 6.2uF +/-10%.


TRX B1+3+7
Co-Pad
Co-Pad
Co-Pad
Co-Pad
Co-Pad
Co-Pad
TRX B2+4
PRX B32
PRX B38/41
TRX B5
TRX B8
TRX B20
TRX B28
TRX B12/17


SFMR8X0C001
ANT_B32
DRX B7
DRX B1/4
DRX B8
DRX B40
DRX B3
DRX B5
DRX B38/41
DRX B2+B4/66 Dual SAW
DRX B20/28
DRX B32
DRX B7+B40+B38/41 LNA
DRX B12/17

49_CONNECTIVITY_CONSYS_MT6631



Note 51-1:	For R3712 size, please select 0402 size or larger one
Note 51-2:	Please refer to MT6765 Baseband design notice for VCN33 LDO selection guide
Note 51-3:	If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND
Note 51-4:	Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support

Title	49_CONNECTIVITY_CONSYS_MT6631			
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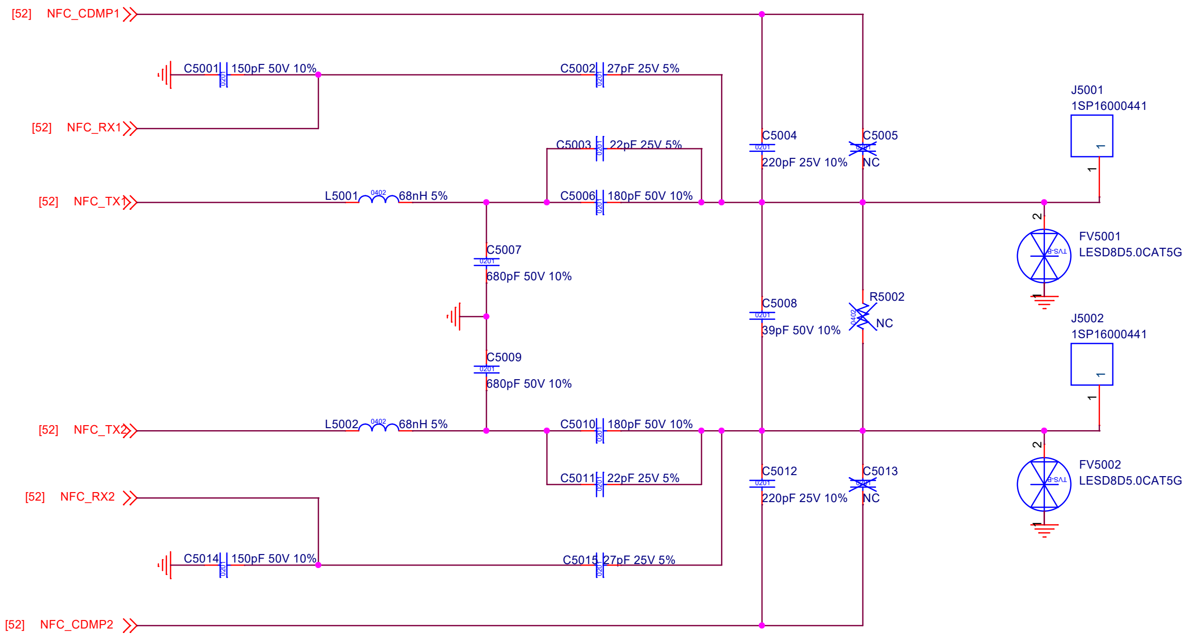
50_ANT_NFC



Title	50_ANT_NFC	
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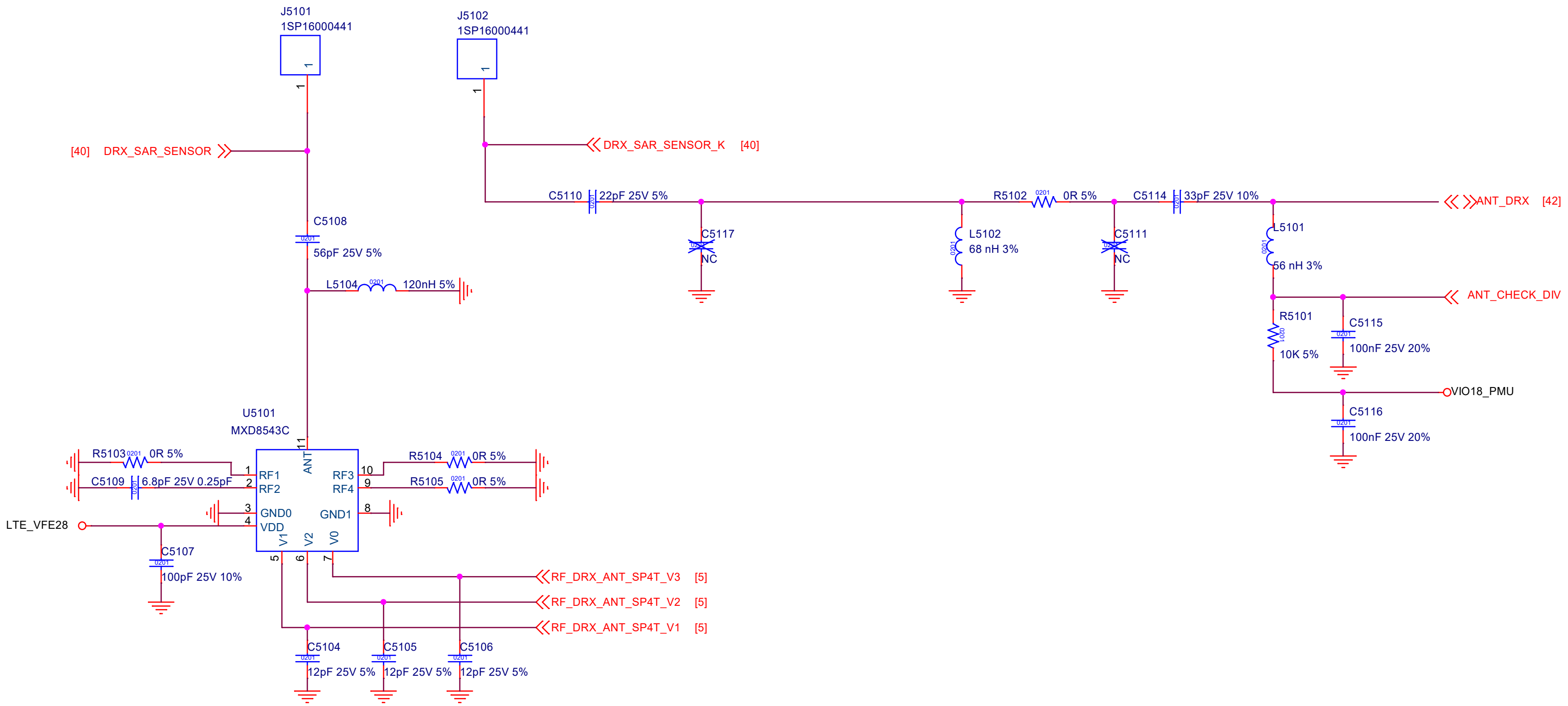
51_ANT_DIV



Title 51_ANT_DIV		
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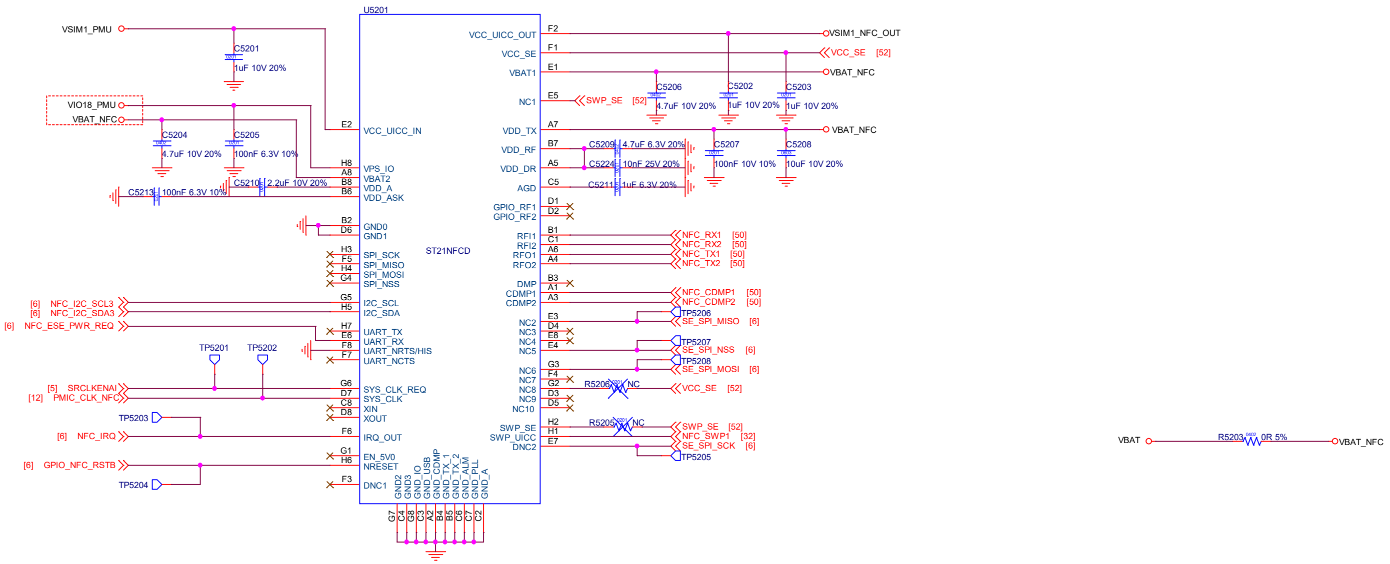
52_NFC



Title	52_ANT_GPS_WIFI	
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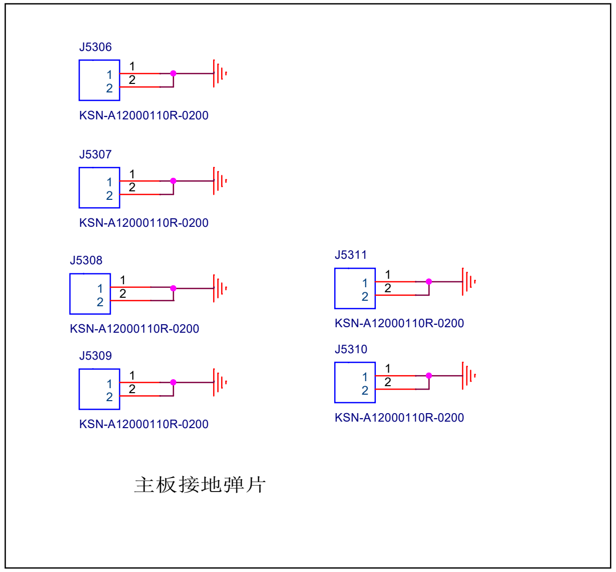
53_ANT_TOP_GND



Title	53_ANT_TOP_GND	
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